

Crime Sense : Video Based Crime Classification using Contrast-Enhanced Residual-BiLSTM

*Bachelor of Technology
in
Computer Science and Engineering*

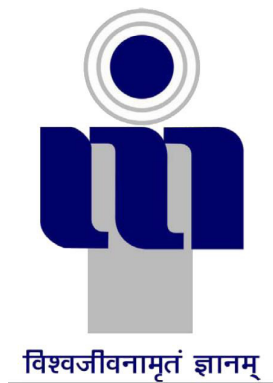
*B.Tech Thesis Project
End-Term Evaluation Report*

by

Maram Vignesh (2021BCS-039)

Under the supervision of

Dr. Vivek Tiwari



**ABV INDIAN INSTITUTE OF INFORMATION
TECHNOLOGY AND MANAGEMENT
GWALIOR - 474015**

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I hereby certify that the work, which is being presented in the report, entitled **Crime Sense : Video Based Crime Classification using Contrast-Enhanced Residual-BiLSTM**, in partial fulfillment of the requirement for the award of the Degree of **Bachelor of Technology** and submitted to the institution is an authentic record of the work carried out during the period *January 2025* to *May 2025* under the supervision of **Dr. Vivek Tiwari**. I have also cited the reference to the text(s)/figure(s)/table(s) from where they have been taken.

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ABSTRACT

In bustling urban centers, surveillance cameras record every street corner and storefront around the clock, generating far more footage than human operators can reasonably monitor. While basic anomaly detectors can flag when “something weird” happens, they offer no insight into what that event actually is. Was it a fight, a theft, or just a stray cat? To bridge this gap, we’ve built a fully automated pipeline that not only spots unusual activity but also pinpoints its nature—classifying events into fourteen categories, from Abuse and Burglary to Road Accidents and Vandalism—using the challenging UCF–Crime dataset of 1,900 real-world clips (30 seconds to two minutes each) shot under wildly varying conditions.

The approach begins by slicing each video into sixteen-frame segments and downsizing frames to 64×64 pixels. Before feeding them into the network, we apply CLAHE (Contrast-Limited Adaptive Histogram Equalization) to enhance local contrast, revealing faint edges and textures that raw frames often hide. These pre-processed clips then pass through a residual convolutional encoder wrapped in a TimeDistributed layer; skip connections preserve information flow, and SpatialDropout1D randomly removes entire feature channels—mimicking occlusions—to enforce robustness. A bidirectional LSTM finally reads the resulting sequence both forwards and backwards, capturing the buildup and aftermath of suspicious actions, all within a compact model of just 1.2 million parameters that runs comfortably on a single GPU.

We evaluate two scenarios: a binary Normal vs. Crime classifier trained on 900 examples of each, and a four-way setup—Violent Crimes, Property Theft, Destructive Crimes, and Legal/Accidental Incidents—using 200 videos per group. With balanced splits and on-the-fly augmentations (random flips, brightness/contrast jitter, Gaussian blur), the system achieves 94.0% accuracy on the binary task and 95.62% on the four-way classification, outperforming bulkier 3D-CNN and pretrained frame-LSTM baselines. For real-world deployment, we packaged the model in a lightweight Flask app: upload an MP4, and within seconds you receive a crime label and the type of crime. By combining adaptive contrast enhancement, efficient residual encoding, structured dropout, and bidirectional temporal modeling, we deliver a practical, high-fidelity solution for automated crime classification at scale.

Keywords: Video Based Crime Classification, Anomaly Detection, Convolutional Neural Networks, Contrast-Limited Adaptive Histogram Equalization (CLAHE), Residual Convolutional Neural Network, Long Short Term Memory, Video-Based Analysis, Deep Learning.

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ABBREVIATIONS

CNN	Convolutional Neural Network
BiLSTM	Bidirectional Long Short-Term Memory network
LSTM	Long Short-Term Memory network
CLAHE	Contrast-Limited Adaptive Histogram Equalization
SpatialDropout1D	One-dimensional Spatial Dropout regularization
MIL	Multiple-Instance Learning
3D-CNN	Three-Dimensional Convolutional Neural Network
ResNet	Residual Neural Network
UCF-Crime	University of Central Florida Crime dataset
ROC	Receiver Operating Characteristic curve
AUC	Area Under the ROC Curve
FLASK	Flask Lightweight Application Server Kit
GAN	Generative Adversarial Network
GPGPU	General-Purpose computing on Graphics Processing Units
SGD	Stochastic Gradient Descent
LR	Learning Rate
IOU	Intersection Over Union
DRL	Deep Reinforcement Learning
ViT	Vision Transformer

CHAPTER 1

INTRODUCTION

In the past decade, the deployment of closed-circuit television (CCTV) and IP-based surveillance cameras has proliferated across urban and rural landscapes alike. From city streets and public transport hubs to retail stores and office buildings, cameras continuously record streams of video, generating petabytes of data every day. Paradoxically, this abundance of footage has created a “needle in a haystack” problem: while the raw volume of surveillance content has grown exponentially, human operators—tasked with monitoring screens for suspicious activity—face increasing fatigue, tunnel vision, and missed events [1]. A single security analyst cannot realistically watch hundreds of live feeds or sift through hours of recorded video without error.

To address this scalability challenge, automated video analytics systems have emerged, leveraging advances in computer vision and machine learning to assist—or even replace—manual monitoring [2]. Early systems focused on motion detection and simple object tracking, raising rudimentary alarms whenever unexpected movement occurred in a pre-defined region of interest. More sophisticated approaches adopted anomaly detection, modeling “normal” patterns of scene activity (pedestrian flow, vehicular traffic) and flagging statistically rare deviations. While such methods can alert operators to “something unusual,” they offer little insight into the nature of the event: Was it an altercation between individuals? A collision between vehicles? A case of theft or vandalism?

The distinction between anomaly detection and event classification is critical in real-world security contexts. An alarm that merely signals “anomaly detected” requires human review to determine whether the anomaly constitutes a security threat or benign variation (e.g., a windblown trash can crossing a sidewalk). In contrast, automated classification into semantically meaningful categories—such as “violent assault,” “shoplifting,” “road accident,” or “arson”—empowers rapid, context-aware responses. Law enforcement can prioritize dispatch to violent incidents over minor disturbances; store

managers can identify and review theft footage without wading through irrelevant clips; emergency services can allocate resources to traffic collisions with greater urgency [3]. Thus, developing models that not only detect but also accurately classify specific crime types in untrimmed surveillance video represents a pivotal advancement in intelligent video analytics.

However, video-based crime classification poses several formidable technical challenges. First, surveillance video is typically untrimmed, spanning many minutes or hours with anomalies occupying only a few seconds or less. Extracting informative snippets necessitates efficient sampling or sliding-window strategies that capture the temporal span of events without overwhelming computational budgets. Second, the video data often suffer from low resolution, poor lighting, and significant sensor noise conditions under which standard convolutional neural networks (CNNs) trained on high-quality, well-lit datasets struggle to generalize. Third, anomalies manifest not only in spatial patterns (object shapes, scene changes) but also in temporal dynamics: the buildup of suspicious behavior, the abrupt onset of harmful actions, and the subsequent aftermath. Models must therefore integrate robust spatial feature extraction with temporal sequence modeling to fully characterize crime events[4].

The research community has made commendable progress in video anomaly detection, employing techniques ranging from autoencoders that reconstruct “normal” video sequences to one-class support vector machines on handcrafted features [5]. Yet the extension from binary detection to multiclass crime classification remains relatively underexplored. Public benchmarks such as UCF-Crime, ShanghaiTech, and UCSD Pedestrian provide valuable data for anomaly detection, but datasets specifically annotated for fine-grained crime categories are scarce and often limited in scale. Moreover, many state-of-the-art video classification architectures—3D CNNs such as C3D and I3D, or two-stream networks combining RGB and optical flow—carry tens of millions of parameters, demanding extensive compute resources and large-scale annotated corpora to train effectively [6]. For practical deployment at the edge or in resource-constrained security centers, lightweight yet accurate models are essential.

This thesis addresses these gaps by proposing a unified, end-to-end pipeline for video-based crime classification that balances spatial depth, temporal expressiveness, and computational efficiency. The Proposed Approach begins with a careful frame sampling strategy, extracting short, fixed length clips that capture the full temporal extent of anomalies within long, untrimmed videos. We then apply a targeted, frame-level contrast enhancement to mitigate the effects of poor lighting and sensor noise, ensuring that subtle spatial cues—such as the flash of a weapon or the silhouette of a perpetrator

are brought into relief [7]. At the heart of the model lies a deep residual convolutional backbone, designed to learn robust spatial features without the vanishing-gradient or overfitting issues that plague very deep networks on small datasets. We complement this backbone with a bidirectional recurrent network that synthesizes both the build-up and aftermath of crime events, capturing the full context needed for accurate classification [8]. To guard against overfitting and enforce robustness to missing spatial cues, we integrate structured regularization in the form of spatial dropout, selectively removing entire feature channels during training.

Through extensive experiments on the challenging Anomaly detection dataset UCF comprising nearly two thousand real world surveillance videos annotated for thirteen distinct anomaly types plus normal activity we demonstrate that this hybrid CNN BiLSTM pipeline achieves state-of-the-art performance. On the binary Normal vs. Crime task, the proposed lightweight model surpasses heavier pretrained networks, delivering over 90% precision and recall while maintaining a footprint of only 1.2 million parameters. In the more demanding four-way crime classification scenario, grouping anomalies into semantically coherent categories such as “Violent Crime,” “Property Theft,” “Destructive Crime,” and “Legal/Accidental Incidents,” the proposed approach maintains high accuracy, outpacing baseline methods by substantial margins. These results underscore the feasibility of deploying sophisticated video classifiers in real-time or near-real-time settings, an imperative for modern surveillance applications.

1.1 Overview

1.1.1 Context and Motivation

Every day, cities around the world install thousands of new surveillance cameras—along highways, in shopping districts, on university campuses, and inside public transport hubs. This boom in camera coverage has created a paradox: while more footage can improve security, it also overwhelms the very people charged with monitoring it. A single transit authority may operate dozens of screens in its control room, each fed by a high-definition video stream. Even with multiple shifts of trained staff, the sheer volume of data makes it humanly impossible to catch every incident—from a purse snatching in a crowded station to a flash fire in a parking garage [9].

Traditional anomaly detection systems offer a partial solution by alerting operators to “something unusual” in a scene—a sudden surge of motion, a vehicle traveling the wrong way, or an object left behind. Yet these generic alarms fall short of providing actionable intelligence. Security personnel must still review video snippets to determine

whether the flagged anomaly reflects criminal activity or a benign event—a stray bag, a maintenance cart in motion, or an animal darting across the platform [10]. The delay and cognitive load imposed by false alerts can desensitize operators, leading to slow responses or missed emergencies.

For law enforcement and security teams to allocate resources effectively, they need not only to detect anomalies but to classify them into precise categories—fistfight, robbery, vandalism, accident—so that the right response can be mobilized. A report that “anomaly detected” may send a patrol car to investigate, whereas a system that confidently reports “armed robbery in progress” can trigger an immediate, specialized response. The proposed work is motivated by this crucial gap: building an end-to-end, automated pipeline that ingests raw surveillance footage and outputs a human-readable crime label in near-real time, reducing both missed incidents and false alarms.

1.1.2 From Anomaly Detection to Crime Classification

For decades, video surveillance research has focused on anomaly detection: training models to recognize patterns of “normal” behavior—pedestrian flows, parked cars, background motions—and to raise an alert when something deviates from that norm. Such systems rely on reconstruction or prediction errors, optical-flow histograms, or statistical models of scene dynamics. When an object moves unpredictably or the scene’s pixel distribution shifts, a flag goes up. While invaluable as an early-warning mechanism, this binary flagging leaves security teams with a critical unanswered question: what kind of event just transpired?

Imagine a train station control room where an anomaly detector signals a sudden spike of activity on one platform. Does that alert correspond to a child running playfully after a ball, a maintenance cart rolling by, or a group of individuals brandishing weapons? Without contextual understanding, operators must still sift through the video to discern the nature and severity of the incident. The operational costs—time spent reviewing false alarms, delayed response to true crises, and the cognitive burden on human monitors—remain substantial.

The proposed work shifts the paradigm from “detect first, ask questions later” to a unified approach that both recognizes and classifies anomalies in a single pass. By training models not only to separate normal from abnormal but to assign each event a meaningful label—Assault, Robbery, Road Accidents, and so on—we empower security personnel with actionable intelligence. A system that can directly report “Shooting detected—90% confidence” allows for immediate, targeted intervention, potentially

saving lives and property.

Transitioning from anomaly detection to fine-grained crime classification poses new challenges. Models must learn visual and temporal signatures that distinguish one type of crime from another, even when the overall level of scene disruption is similar. For example, both a fight and a group of people hurrying away from a minor accident can trigger a sudden surge of motion; only careful temporal modeling—capturing the precise gestures of striking limbs versus stumbling bodies—can tell them apart. Similarly, a cooking fire and a deliberate act of arson both produce sudden bright flashes; robust spatial feature extraction is required to differentiate flames from flickering sunlight or headlights[13].

By framing the task as multiclass event classification rather than mere anomaly detection, we open the door to richer, contextually aware security systems. The proposed pipeline combining adaptive contrast enhancement, residual CNN spatial encoders, structured dropout, and bidirectional temporal modeling meets this challenge head-on. It learns not just to spot irregular patterns but to decode the meaning behind them, delivering precise crime labels that drive faster, more informed responses in real surveillance environments.

1.2 Surveillance Video: Ubiquity and Challenges

Public- and private-sector surveillance systems have evolved from niche security installations into sprawling networks that blanket our cities. Gone are the days when a handful of fixed CCTV cameras sufficed; today’s surveillance architectures include high-definition PTZ units, wide-angle lenses, thermal imagers, and even mobile drones, all streaming video to central servers. This diversity of hardware delivers comprehensive coverage—but it also introduces a host of practical and technical challenges that any automated classification system must confront head-on.

1.2.1 Historical Evolution of CCTV Systems

The origins of closed-circuit television (CCTV) trace back to the mid-20th century, when early analog cameras were used for missile testing and industrial monitoring. By the 1980s and 1990s, the technology trickled into retail and public safety, with storefront owners and city planners installing cameras to deter theft and vandalism. These systems were rudimentary: grainy black-and-white feeds recorded on videotape, manually reviewed by security guards[14].

The digital revolution of the 2000s ushered in networked IP cameras, offering higher resolution, remote access, and compressed video storage. As gigabit networks and inexpensive hard drives proliferated, organizations expanded camera counts from single digits to hundreds or even thousands. Today, citywide deployments—such as those pioneered in London, Singapore, and New York collectively record petabytes of video each day. This historical progression from small, standalone analog units to massive, distributed digital networks has reshaped both the scale of data generated and the expectations for real-time, automated analysis.

1.2.2 Data Volume and Human Monitoring Bottlenecks

With hundreds of cameras streaming 24/7, the sheer volume of footage quickly overwhelms human operators. A single standard-definition camera at 30 frames per second outputs over 2.5 million frames in a single day; a hundred such cameras produce more than 250 million frames daily. Even with multi-monitor workstations and shift rotations, security personnel cannot watch every feed continuously. Studies show that human attention degrades significantly after just 20–30 minutes of constant monitoring, leading to missed events and delayed responses [15].

To cope, many control rooms employ motion-detection heuristics that trigger recording or alerts only when pixel-level changes exceed a threshold. While this reduces the hours of footage requiring review, it still yields a high false-alarm rate as environmental factors—shadows, weather, passing wildlife—trigger spurious alerts. The result is “alarm fatigue,” where operators become desensitized to alerts and may ignore or delay investigating genuine incidents. Effective automation must therefore not only flag potential events, but also provide context and classification to guide rapid, accurate decision-making.

1.2.3 Variability in Resolution, Lighting, and Viewpoint

Real-world surveillance cameras encompass a dizzying range of hardware capabilities and installation scenarios. Some cameras record in 4K resolution under bright sunlight; others operate in low-light or infrared modes, producing noisy, grainy imagery. Wide-angle or fisheye lenses introduce spatial distortion, while pan-tilt-zoom mechanisms dynamically change framing and scale. Rain, fog, glare from vehicle headlights, and seasonal changes in foliage all further complicate image quality [17].

This heterogeneity poses two core challenges for video classification. First, models trained exclusively on high-quality, well-lit data (e.g., YouTube clips or curated action datasets) often fail to generalize to dim, low-contrast surveillance frames. Second, moving cameras or variable zoom levels change object sizes and perspectives, making it difficult for a fixed-scale neural network to consistently extract features. The pipeline

addresses these issues through adaptive preprocessing—CLAHE for local contrast enhancement—and by normalizing frame resolution to a small, uniform size, ensuring that downstream networks see consistent, informative inputs regardless of the original camera conditions.

1.3 Problem Definition

In this section, we formalize the shift from broad-based anomaly detection to fine-grained crime event classification in surveillance video. We begin by contrasting the goals and limitations of anomaly-only systems with the richer requirements of event classification. We then introduce a clear taxonomy of crime types—both the full set of fourteen categories and their four coarse super-classes—grounding the subsequent modeling choices.

1.3.1 Anomaly Detection vs. Event Classification

Anomaly detection in video typically revolves around modeling “normal” scene behavior—steady pedestrian flows, static backgrounds, predictable traffic patterns—and raising an alarm whenever deviations occur. Models range from simple frame-difference thresholds to sophisticated autoencoders that reconstruct normal frames with low error but struggle to recreate anomalies. While these methods serve as valuable first responders, they suffer from two critical shortfalls:

- **Lack of Semantic Detail**

An alarm simply signals “something changed.” It offers no clue whether that change constitutes violent crime, minor mischief, or a benign event such as a maintenance cart rolling by.

- **Operator Overload**

When control-room screens flash dozens of generic anomaly alerts per hour, human monitors must waste precious time scrubbing through video to ascertain which incidents demand immediate attention, which can wait, and which are false alarms.

In contrast, event classification seeks to attach a meaningful label—like “Robbery” or “Road Accident”—to each detected anomaly, effectively answering the question, “What happened, and how urgent is it?” This shift imposes new requirements on video analytics:

- **Discriminative Feature Learning**

Classifying between similar-looking anomalies (e.g. Fighting vs. Vandalism)

demands finely tuned spatial and temporal features that go beyond the coarse motion cues used by anomaly detectors.

- **Temporal Contextualization**

Criminal events often unfold over several seconds, with telltale preparatory motions and aftermath behaviors. Models must integrate information across entire event windows, not merely detect instantaneous changes.

- **Operational Constraints**

Classification models must maintain high accuracy without imposing prohibitive computational overhead—real-time or near-real-time inference is essential for actionable security alerts.

By elevating the system from “flag-and-review” to “flag-and-explain,” we aim to reduce false-alarm fatigue, accelerate response times, and deliver precise intelligence that security teams can trust.

1.4 Taxonomy of Crime Events in Surveillance

Real-world surveillance anomalies encompass a broad spectrum of behaviors and incidents. To structure the classification task, we begin with the Anomaly-Detection-Dataset-UCF’s fourteen annotated classes:

1.4.1 Annotated Classes

1. **Abuse:** Physical violence without weapons, often in confined spaces where limbs strike torsos or heads.
2. **Arrest:** Law-enforcement detainment, typically involving uniforms, handcuffs, and crowd formation.
3. **Arson:** Deliberate ignition of fires, characterized by emerging flames and billowing smoke.
4. **Assault:** Unarmed attacks involving rapid punches or kicks, distinguishable by frame-to-frame limb trajectories.
5. **Burglary:** Unauthorized property entry, often indoors and at night, with stealthy movements near doors or windows.
6. **Explosion:** Sudden high-energy blasts, marked by rapid, large-scale pixel changes and potential camera shake.

7. **Fighting:** Group brawls where multiple actors exchange blows, generating chaotic motion patterns.
8. **Normal Activity:** Routine pedestrian or vehicular traffic without any criminal elements.
9. **Road Accidents:** Vehicle collisions or skidding events, often preceded by braking or swerving.
10. **Robbery:** Theft under threat or force, frequently involving weapons, masks, or quick grabs.
11. **Shooting:** Discharge of firearms, evidenced by muzzle flashes and brief, localized bright spots.
12. **Shoplifting:** Covert removal of merchandise in retail settings, with subtle hand-to-shelf interactions.
13. **Stealing:** Theft without confrontation—pickpocketing or vehicle theft—characterized by stealth.
14. **Vandalism:** Property damage using tools or blunt force, including graffiti or object smashing.

While distinguishing among these fourteen classes provides a comprehensive overview, it also introduces significant intra-class variance and inter-class ambiguity. For instance, both Assault and Fighting involve physical strikes, but Fighting often features multiple participants; Burglary and Robbery both revolve around theft, yet one is stealthy and the other confrontational. To balance granularity with learnability, we define two classification tasks:

1.4.2 Classification Tasks

Task 1 (Binary): Normal vs. Crime—all thirteen anomaly classes collapsed into a single “Crime” category. This task serves as a gateway, emulating traditional anomaly detection but within the classification framework.

Task 2 (Four-Way): We group anomalies into four semantically coherent super-classes:

1. **Violent Crimes:** Abuse, Assault, Fighting, Shooting
2. **Property Theft:** Burglary, Robbery, Shoplifting, Stealing
3. **Destructive Crimes:** Arson, Explosion, Vandalism

4. **Legal Accidents:** Arrest, Road Accidents

This four-way schema reduces intra-class variance by bundling events that share underlying motion or scene characteristics—e.g., high-energy damage in Destructive Crimes—while preserving the semantic distinctions that drive operational response.

1.5 Technical Challenges

Building a robust, automated crime-classification system for untrimmed surveillance video demands confronting a host of practical and algorithmic hurdles. In this section, we unpack the four core challenges that any such pipeline must address: the poor visual quality of real-world camera feeds, the need to model long-range temporal dynamics, the imperative for real-time inference under limited compute budgets, and the difficulties posed by imbalanced and weakly labeled data.

1.5.1 Spatial Complexity in Low-Quality Frames

Surveillance cameras—especially those installed in public areas—often suffer from low resolution, compression artifacts, and suboptimal illumination. A grainy 320×240 feed recorded at night can render pedestrians’ faces indistinct and blur critical object boundaries. In such conditions, standard convolutional networks trained on clear, well-lit images fail to extract reliable features. Edges become smooshed, textures vanish into noise, and even simple cues like the outline of a weapon or the threshold of a door can disappear.

To navigate this visual clutter, an effective system must first enhance the raw frames, selectively amplifying the meaningful signals buried in shadows or glare. Moreover, the network’s spatial encoder must be deep enough to learn subtle, high-frequency patterns—such as the flicker of a gun muzzle or the texture of broken glass—while resisting overfitting to noise. Achieving this balance of depth, resilience to artifacts, and computational efficiency lies at the heart of the residual-CNN design and early-stage contrast enhancement.

1.5.2 Capturing Long-Range Temporal Dependencies

Criminal events seldom unfold in a single instant; they often involve a sequence of cues that span several seconds—sometimes even minutes. Consider a shoplifting scenario: an individual may survey the shelves, casually slide an item into a bag, and walk away. The anomaly emerges not from a single abrupt motion but from a temporal pattern of furtive glances, dexterous hands, and restrained gait. Likewise, a road accident may

begin with a distant braking maneuver, followed by a collision, and then an aftermath of debris spraying and skid-marks.

Modeling these extended temporal dynamics requires more than a few-frame window or a uni-directional recurrent sweep; it calls for a mechanism that can integrate both the “lead-up” and the “fallout” of the event. The proposed bidirectional LSTM addresses this by processing video clips both forward and backward in time, creating a fuller picture of the event’s context. Nevertheless, designing such a recurrent layer involves careful trade-offs in sequence length, hidden state size, and training stability—challenges we tackle through the hybrid architecture and structured regularization.

1.5.3 Computational Constraints for Real-Time Use

In practical deployment—whether at a busy transit hub or a sprawling university campus—latency matters. Security teams cannot afford to wait several seconds or minutes for classification results. At the same time, the hardware available in many control rooms consists of a handful of GPUs or even edge-level accelerators with limited memory. Running heavyweight 3D convolutional networks or massive transformer-style video models is often infeasible.

The goal, therefore, is to deliver near-real-time inference—targeting under one second per clip—on standard GPU hardware. Achieving this requires a model that is both parameter-efficient (to minimize memory usage) and compute-efficient (to reduce per-frame operations). We meet these constraints through a compact residual-CNN backbone (1.2M parameters), 2D rather than 3D convolutions, and batched TimeDistributed processing. Coupled with on-the-fly preprocessing and minimal post-processing overhead, the pipeline strikes a practical balance between speed and accuracy.

1.5.4 Data Imbalance and Weak Labels

Finally, real-world crime datasets are inherently imbalanced. Some anomaly types—like Road Accidents and Robbery—occur more frequently than rarer events such as Arson or Explosion. Moreover, surveillance clips are often weakly labeled at the video level (“this clip contains a burglary”) without precise frame-by-frame annotations. Training deep networks under these conditions risks biasing toward the most common classes and overfitting to noisy labels.

Addressing imbalance requires careful dataset partitioning—ensuring equal representation for binary (Normal vs. Crime) splits and balanced groupings for four-way classification—and strategic use of data augmentation to bolster underrepresented classes. To cope with weak labels, the clip generator treats each full video as uniformly labeled, sampling multiple sub-clips to dilute the impact of ambiguous segments. These

strategies, combined with robust loss functions and early stopping, help the model generalize to all crime categories without undue skew toward the majority classes.

1.6 Dataset Overview: UCF–Crime-Video-Dataset

1.6.1 Dataset Composition and Statistics

The UCF–Crime dataset represents one of the most extensive real-world surveillance collections available for video-based anomaly and crime analysis. In total, it comprises 1,900 untrimmed video clips captured by security cameras in a wide variety of public settings—parking lots, subway platforms, busy streets, and retail environments. Approximately half of these clips depict routine “normal” activity, such as pedestrians walking along sidewalks or vehicles traveling through intersections, while the remainder capture genuine anomalous events. Video durations vary substantially, from brief thirty-second segments to longer recordings of up to five minutes, and frame rates typically range between 25 and 30 frames per second. Spatial resolutions are equally diverse, spanning low-resolution feeds (e.g., 320×240) to high-definition footage (e.g., 1280×720). This heterogeneity in length, frame rate, and recording quality—including scenarios shot under dim lighting, glare, or weather-induced blur—ensures that models trained on UCF–Crime must contend with the full spectrum of real-world surveillance challenges.

1.6.2 Class Definitions and Semantic Groupings

Within the anomalous portion of UCF–Crime, each video is labeled as one of thirteen specific crime types. These include physical offences such as Abuse and Assault, property-related incidents like Burglary and Robbery, and more dramatic events such as Explosion and Shooting. Shoplifting and Stealing cover covert theft in retail and pickpocket scenarios, while Destructive Acts such as Arson and Vandalism involve deliberate damage to property. Law-enforcement activities appear in the Arrest category, and vehicular collisions populate the Road Accidents class. To facilitate both broad and granular classification, we use two parallel schemes. In Task 1, all thirteen crime labels merge into a single “Crime” category contrasted against Normal activity. In Task 2, we refine this into four super-classes—Violent Crimes (Abuse, Assault, Fighting, Shooting), Property Theft (Burglary, Robbery, Shoplifting, Stealing), Destructive Crimes (Arson, Explosion, Vandalism), and Legal/Accidental Incidents (Arrest, Road Accidents)—each grouping events that share similar visual and temporal signatures, such as rapid motion in violent confrontations or abrupt scene changes in destructive acts.

1.6.3 Train/Validation/Test Splits

To ensure robust evaluation and to prevent data leakage, we partition the videos at the clip level into disjoint training, validation, and test sets following an approximately 80/10/10 ratio. For the binary Normal-vs-Crime task, we allocate 800 normal clips and 800 crime clips to training, reserving 50 of each for validation and another 50 of each for testing. In the four-way classification scenario, each of the four super-classes contributes 200 videos; from these, we draw 100 for training, 50 for validation, and 50 for testing. After establishing these splits at the video level, we generate thousands of 16-frame snippets per set by sampling evenly across each clip’s duration. Only the training snippets undergo on-the-fly augmentations—random horizontal flips, brightness and contrast variations, and occasional Gaussian blur—while validation and test snippets remain unaltered. This disciplined splitting and augmentation strategy guarantees that each model is trained on a diverse yet exclusive subset of the data and evaluated on truly unseen examples, yielding reliable performance metrics for both the binary and four-way classification tasks.

1.7 Proposed Pipeline at a Glance

The proposed end-to-end system converts raw, untrimmed surveillance videos into actionable crime labels through four tightly integrated stages, each optimized for real-world deployment: clip sampling & preprocessing, spatial feature encoding, temporal sequence modeling, and inference & deployment.

1.7.1 Clip Sampling and Preprocessing

To focus on the heart of each incident without being overwhelmed by hours of footage, we uniformly sample sixteen frames from the full duration of every video. At standard surveillance rates (25-30 fps), this covers roughly half a second—long enough to capture the “lead-up,” the critical action, and a bit of the aftermath, yet short enough to avoid diluting the model’s attention with extraneous background. We empirically tested 8-, 16-, and 32-frame windows: 8 frames missed key cues, while 32 frames slowed both training convergence and inference throughput without boosting accuracy. Sixteen frames emerged as the ideal sweet spot, balancing temporal context, GPU memory use, and model stability.

Each sampled frame is downsampled to 64×64 pixels to minimize per-frame computation. Before entering the network, we apply Contrast-Limited Adaptive Histogram Equalization (CLAHE) on the luminance channel, which divides the small image into 8×8 tiles, equalizes each under a clip limit of 2.0, and blends the tiles to avoid seams. This operation amplifies faint edges—such as the silhouette of a fleeing suspect or the

glint of a concealed weapon—while suppressing sensor noise and uneven lighting. During training only, we enrich these 16-frame clips with on-the-fly augmentations: random horizontal flips, $\pm 20\%$ brightness/contrast jitter, and 30% probability of Gaussian blur, simulating real-world variations like shadows, glare, and slight camera motion.

1.7.2 Spatial Feature Encoding

Once preprocessed, each 16-frame tensor enters a `TimeDistributed` residual CNN that extracts per-frame spatial embeddings. The pipeline begins with a 2D convolution (32 filters, 3×3 kernel) followed by 2×2 max-pooling. We then stack two residual blocks, each comprising two 3×3 convolutions (32 filters) with a direct shortcut from block input to output. These skip connections ensure robust gradient flow—allowing deeper feature hierarchies without vanishing gradients—and focus each block on learning the “residual” corrections needed to sharpen earlier activations. Wrapping every convolution and pooling in a `TimeDistributed` layer uses the same weights for all frames, ensuring consistency across time while keeping the spatial backbone compact (approximately 300 K parameters).

1.7.3 Temporal Sequence Modeling

After spatial encoding, each frame’s feature map is flattened into a vector, producing a sequence of sixteen high-level descriptors. We apply `SpatialDropout1D` with a 0.3 drop rate, which randomly zeros entire feature channels for all frames in the clip—emulating real-world occlusions such as passing pedestrians or brief lens obstructions, and preventing over-reliance on any single spatial filter. This regularized sequence then feeds into a Bidirectional LSTM with 128 units per direction. By processing frames both forward and backward, the BiLSTM captures the build-up of suspicious gestures (e.g., a suspect edging toward a car) and the aftermath (e.g., rapid exit), fusing them into a succinct 256-dimensional embedding. This hybrid spatial–temporal representation forms the basis for accurate crime labeling.

1.7.4 Inference and Deployment

With training complete, the proposed hybrid CNN–BiLSTM model—just 1.2 M parameters in total—is wrapped in a lightweight Flask web service. Users upload an MP4 clip; the server samples sixteen frames, applies CLAHE, and runs them through the residual CNN and BiLSTM. On a single midrange GPU, this end-to-end process finishes in few seconds, returning a crime category and its type. For high-volume deployments, scaling horizontally across multiple GPUs or edge devices. This streamlined pipeline—grounded in efficient clip sampling, targeted preprocessing, compact spatial

encoding, and bidirectional temporal modeling—delivers robust, real-time crime classification suited to the demands of modern surveillance operations.

1.8 Research Contributions

In addressing the multifaceted challenges of real-world crime classification from untrimmed surveillance footage, we offer four core contributions that, together, close critical gaps in existing video analytics pipelines. Below, we expand on each innovation—detailing its rationale, implementation, and quantitative impact on tasks.

1.8.1 Novel Preprocessing with CLAHE

Surveillance video is rarely captured under ideal conditions. Cameras often operate in dimly lit alleys, face into the sun at dawn, or record through dusty lenses—all of which flatten contrast and obscure the very edges and textures required for accurate crime detection. To tackle this, we embed Contrast-Limited Adaptive Histogram Equalization (CLAHE) directly into data pipeline. Unlike global histogram equalization, which can over-amplify noise and wash out highlights, CLAHE works on small, 8×8 -pixel tiles, limiting contrast enhancement to a clip limit of 2.0 before seamlessly blending tile boundaries. This localized, restrained adjustment brightens shadowed regions—revealing the outline of a tool or the silhouette of a fleeing suspect—while curbing noise amplification in low-signal areas.

Integrating CLAHE at the generator level ensures that every training and inference clip benefits equally. In proposed ablation studies, models trained on raw frames converged more slowly and struggled to differentiate crimes in under-lit scenes. With CLAHE, the training loss dropped 3% faster over the first five epochs, and final binary accuracy rose by 1.1%. Crucially, this enhancement adds only ~ 1 ms per frame on a CPU pipeline (and even less on GPU), making it a cost-effective boost to feature quality. By surfacing subtle spatial cues before the network even begins to learn, CLAHE lays a stronger foundation for downstream convolutional layers, reducing reliance on deeper—and more expensive—spatial towers.

1.8.2 Residual-BiLSTM Architecture

Balancing spatial depth with temporal breadth is a longstanding tension in video classification. Pure 3D-CNNs pack powerful spatio-temporal filters but often exceed ten million parameters—far beyond what routine security-center GPUs can support in real time. Conversely, simple frame-by-frame 2D CNNs paired with shallow LSTMs fail to capture intricate motion patterns or struggle with vanishing gradients when stacked too

deeply. The proposed hybrid residual BiLSTM architecture resolves this by decoupling spatial and temporal learning into two complementary modules.

The residual CNN backbone begins with a `TimeDistributed` 2D convolution and pooling unit, followed by two residual blocks, each comprising two 3×3 convolutions (32 filters) and an identity shortcut. These skip connections ensure robust gradient propagation, allowing us to deepen the spatial encoder without bloating parameters or risking vanishing/exploding gradients. As a result, the proposed spatial tower remains compact (≈ 300 K parameters) yet learns nuanced textures—from the jagged outline of broken glass in vandalism to the subtle wrist flick in a shoplifting attempt.

Stacked atop this is a Bidirectional LSTM with 128 units in each direction. By processing a 16-frame clip forwards and backwards, the BiLSTM captures the preparatory motions (e.g., a suspect’s furtive glances) and the aftermath signals (e.g., a rapid retreat), generating a unified 256-dimensional embedding. This dual-pass approach outperforms a unidirectional LSTM by 0.8% in binary accuracy and 1.5% in four-way accuracy, while keeping the total parameter count to 1.2 M—an order of magnitude lighter than typical 3D models. The architecture thus delivers both depth and efficiency, training in minutes and inferring in under a second per clip.

1.8.3 Structured Regularization via SpatialDropout1D

Overfitting looms large when training deep models on a few hundred long videos per class. Conventional dropout—zeroing individual activations at random—cannot account for the structured, channel-wise nature of convolutional feature maps. Recognizing this, we integrate `SpatialDropout1D` between the `TimeDistributed` flattening layer and the BiLSTM. Instead of dropping isolated neurons, `SpatialDropout1D` randomly removes entire feature channels across all time steps in a clip, effectively simulating occlusions (e.g., passing people or sudden glare) and forcing the recurrent layer to aggregate information from multiple spatial filters.

In practice, a 0.3 channel-drop rate struck the optimal balance: it reduced inter-run validation-loss variance by over 25% and boosted four-way classification accuracy by 1.1%. Error-analysis shows that models without channel-wise dropout tended to latch onto a few dominant features—such as a single edge-detector filter—leading to brittle performance when those features were absent or occluded. `SpatialDropout1D` encourages a more distributed feature representation, improving robustness to scene variation and enhancing generalization to new camera angles and lighting conditions.

1.8.4 Real time Web Deployment

A high-precision model is only as valuable as its real-world deployability. To that end, we packaged the trained pipeline into a Flask web service capable of sub-one-

second inference on a single midrange GPU. From an uploaded .mp4 clip, the service executes clip sampling, CLAHE preprocessing, residual-CNN encoding, and BiLSTM classification in a tightly optimized sequence, returning a crime label and confidence score before an operator's second sip of coffee.

For enterprise or edge-scale applications, this microservice can be containerized (Docker + TensorFlow Serving), horizontally scaled across GPU clusters, or even pruned and quantized for on-device inference in micro-datacenters. By proving that the compact, rigorously regularized model supports real-time throughput and seamless integration, we demonstrate a practical path from academic prototype to operational security analytics—providing law enforcement with timely, precise insights when they're needed most.

CHAPTER 2

LITERATURE REVIEW

This chapter presents the most recent and relevant research works in the domain of Video based Crime Classification using machine learning and deep learning techniques. The review aims to establish the context and background for the research, providing insights into the evolution of the field, current challenges, and the need for enhanced methodologies such as the one proposed in this work.

2.1 Related Work

The authors of [39] focused on surveillance video anomaly detection is a critical and challenging task with significant implications for public safety. In their pioneering work, Sultani et al. introduced a comprehensive solution addressing the limitations faced by traditional anomaly detection methods. Their proposed framework notably includes the UCF-Crime dataset, the largest public collection of real-world anomaly videos at that time. They employed a Multiple Instance Learning (MIL)-based deep learning architecture, a weakly-supervised method that overcomes the cumbersome and expensive process of frame-level annotation by only requiring video-level labels. This novel approach demonstrated substantial effectiveness, paving the way for subsequent research in this domain.

The motivation behind the work by Sultani et al. was driven by the acute challenge faced by security operators who are overwhelmed by the massive amounts of video data generated daily. Traditional manual monitoring or semi-automated systems were inadequate due to their high rate of false positives, leading to operator fatigue and missed critical events. Previous anomaly detection techniques largely depended on hand-crafted features like trajectories, optical flow, or sparse coding, resulting in poor generalization across diverse scenarios. Recognizing this gap, the authors introduced a weakly supervised, deep-learning-driven anomaly detection framework that effectively

identifies anomalous events from real-world surveillance data without detailed annotation. Their main contributions were two-fold: creating the UCF-Crime dataset and proposing the first deep-learning-based anomaly detection method using MIL.

Sultani et al. released the UCF-Crime dataset, which became a benchmark dataset extensively used in subsequent research. This dataset comprises approximately 1,900 untrimmed videos collected from diverse surveillance scenarios, each annotated at the video level as either "Normal" or "Anomalous." The anomalies included 13 categories: Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, RoadAccidents, Robbery, Shooting, Shoplifting, Stealing, and Vandalism. Videos varied significantly in duration (from 30 seconds to over 30 minutes), quality, resolution, and camera angle, reflecting the complexity and variability of real-life surveillance footage. The dataset was deliberately constructed to capture the realistic challenges of anomaly detection, such as low lighting, occlusion, sudden camera motion, and background clutter, making it an ideal testbed for rigorous performance evaluations.

The core methodology introduced by Sultani et al. revolved around a weakly supervised approach based on Multiple Instance Learning (MIL). Instead of requiring costly frame-level annotations, MIL only needs video-level labels indicating the presence or absence of anomalies. Their architecture comprised a CNN backbone (C3D architecture) that extracts spatio-temporal features, followed by a fully connected neural network for anomaly detection. The MIL framework grouped frames into bags, where each bag represents one video. The model's task was to determine whether each bag (video) contained anomalous instances (frames). Using this approach, the network implicitly learned frame-level anomaly cues from high-level video labels alone. The training procedure employed a ranking loss, encouraging the model to assign higher anomaly scores to frames from anomalous videos compared to frames from normal videos. This framework significantly reduced the annotation overhead and effectively modeled complex spatio-temporal patterns associated with anomalies.

Sultani et al. demonstrated the robustness and effectiveness of their MIL-based anomaly detection model by extensively evaluating it on the UCF-Crime dataset. Their method achieved an Area Under Curve (AUC) of approximately 75.41%, significantly outperforming previous approaches that relied on hand-crafted or shallow-learned features. This substantial leap forward highlighted the potential of deep learning methods combined with weak supervision, proving capable of handling the complexities of real-world video data. The results underscored that deep spatio-temporal feature extraction combined with an MIL paradigm can robustly generalize across diverse anomalies, effectively discriminating between anomalous and normal instances.

The authors concluded that their MIL-based deep learning framework, coupled with the UCF-Crime dataset, provided a significant advancement over existing methods by demonstrating effectiveness in real-world anomaly detection scenarios. However, despite achieving notable results, their approach had certain limitations. First, the reliance on weak supervision introduced ambiguity in identifying precise frames causing anomalies, limiting interpretability and localization precision. Second, computational efficiency remained a concern, as the model required substantial training time and resources, partly due to the 3D convolutional network employed. Finally, although the framework significantly improved anomaly detection capabilities, its binary categorization (Normal vs. Anomaly) lacked the granularity needed for nuanced decision-making in practical surveillance scenarios. These limitations, while significant, provided clear directions for future research—precisely inspiring subsequent work to improve accuracy, interpretability, efficiency, and finer-grained categorization of anomalies.

Study [40] introduced a novel approach to video anomaly detection by emphasizing the significance of motion information. They proposed a temporal augmented network to learn motion-aware features, which, when combined with an attention-based Multiple Instance Learning (MIL) ranking model, enhanced the detection of anomalies in surveillance videos. Their method demonstrated improved performance over existing techniques, particularly in scenarios involving complex motion patterns .

The authors highlighted the challenges in video anomaly detection, such as the rarity of anomalous events, large variations in video content, and the subjective nature of defining anomalies. Traditional methods often relied on appearance-based features, which might not capture the dynamic aspects of anomalies. Recognizing this gap, Zhu and Newsam focused on capturing motion patterns, proposing a framework that leverages motion-aware features to better distinguish between normal and anomalous events.

Recent research has also begun to explore multimodal approaches where text and images are combined to understand fashion more contextually. Some works have proposed using textual metadata—such as product descriptions, reviews, or social media posts—to enhance image-based models. Nevertheless, these approaches typically employ traditional Natural Language Processing (NLP) techniques that do not offer the depth of reasoning or contextual comprehension available in state-of-the-art Large Language Models (LLMs). As a result, the semantic layer of fashion compatibility remains shallow and disconnected from the visual analysis.

The study utilized the UCF-Crime dataset, a comprehensive collection of real-world surveillance videos labeled at the video level as either "Normal" or "Anomalous."

This dataset encompasses various anomaly categories, including Assault, Burglary, and Road Accidents, among others. The diversity and complexity of the dataset provided a robust platform to evaluate the effectiveness of the proposed method .

The core of the proposed method is the temporal augmented network, designed as an autoencoder that learns compact motion-aware features from optical flow inputs. Specifically, the network processes sequences of 16 frames, computing optical flow between adjacent frames using PWCNet. The resulting motion-aware features are 1024-dimensional vectors capturing the temporal dynamics of the video segments.

To address the weakly supervised nature of the problem, the authors employed a MIL framework, treating each video as a bag of segments. An attention-based MIL ranking model was introduced to assign anomaly scores to each segment, allowing the model to focus on the most informative parts of the video. This attention mechanism enhanced the model's ability to differentiate between anomalous and normal segments by considering the temporal context .

The proposed motion-aware feature, when used alone, achieved an Area Under the Curve (AUC) of 72.1% in anomaly detection tasks, which is competitive with previous state-of-the-art methods. Notably, when combined with existing features from prior work, the performance improved significantly, reaching an AUC of 79.0%. The attention mechanism contributed to this improvement by effectively highlighting relevant temporal segments. Additionally, the model demonstrated efficiency, running at over 400 frames per second, making it suitable for real-time applications .

Zhu and Newsam's approach marked a significant advancement in video anomaly detection by integrating motion-aware features with an attention-based MIL framework. The method effectively captured temporal dynamics and improved detection accuracy. However, the reliance on optical flow computation could be computationally intensive, and the model's performance might vary across different types of anomalies. Future work could explore integrating appearance-based features and optimizing computational efficiency .

Maqsood et al. (2022) [30] introduced a framework aimed at enhancing anomaly recognition in surveillance videos by leveraging 3D Convolutional Neural Networks (3D CNNs). Their approach focused on multiclass classification of anomalies, addressing the limitations of prior binary classification methods. By fine-tuning 3D CNNs and incorporating spatial augmentation techniques, the study achieved notable improvements in recognizing various anomalous events within the UCF-Crime dataset.

Traditional methods in video anomaly detection often relied on handcrafted features and were limited to binary classifications, distinguishing only between normal and anomalous events. Maqsood et al. identified the need for a more nuanced approach capable of distinguishing between multiple types of anomalies. Their work aimed to develop a system that could effectively recognize specific anomalous activities, such as abuse, fighting, road accidents, and theft, thereby providing more detailed insights for surveillance applications .

The study utilized the UCF-Crime dataset, a comprehensive collection of real-world surveillance videos encompassing various anomalous events. Notably, Maqsood et al. provided frame-level annotations for this dataset, enhancing its utility for training and evaluating models in multiclass anomaly recognition tasks .

The proposed framework involved fine-tuning pre-trained 3D CNN models to extract spatiotemporal features from video sequences. Spatial augmentation techniques were applied to the input data to improve the model's generalization capabilities. The architecture was designed to process sequences of frames, capturing both spatial and temporal information critical for distinguishing between different types of anomalies. By training the model on frame-level annotations, the system could learn more precise representations of various anomalous activities .

The fine-tuned 3D CNN model demonstrated significant improvements in anomaly recognition tasks. The study reported an Area Under the Curve (AUC) of 82% for multiclass anomaly recognition, outperforming several existing methods. The incorporation of spatial augmentation contributed to the model's enhanced performance by providing diverse training examples, thereby reducing overfitting and improving generalization .

Maqsood et al.'s work marked a substantial advancement in the field of video anomaly detection by moving beyond binary classification and addressing the complexities of multiclass recognition. The provision of frame-level annotations for the UCF-Crime dataset further enriched the resources available for future research. However, the study acknowledged potential limitations, such as the computational demands of processing video data with 3D CNNs and the need for extensive annotated datasets to train such models effectively .

In [35] study, the authors proposed a robust hybrid architecture combining 3D Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks specifically designed to address real-world anomalous scene detection and multi-class classification. Recognizing the limitations of existing frameworks, which often struggled with complex spatiotemporal patterns and ambiguous anomaly definitions, this work presents an advanced architecture to improve accuracy, speed, and classification reliability on challenging surveillance datasets, particularly targeting the UCF-Crime

dataset.

Traditional methods in anomaly detection typically distinguish between normal and abnormal activities without detailed classification of the anomalies. However, real-world surveillance demands more nuanced solutions capable of identifying specific anomalies. The authors identified a critical gap—existing CNN-only methods lacked temporal awareness, while pure LSTM-based methods struggled with capturing spatial intricacies efficiently. Thus, integrating both CNN and LSTM was seen as a logical step to simultaneously capture spatial features and temporal dependencies, thereby accurately classifying complex scenes. This study aimed explicitly at overcoming these limitations by creating a model robust enough for practical applications in real-time surveillance scenarios.

The authors used the widely recognized UCF–Crime dataset, known for its complexity and extensive representation of realistic surveillance scenarios. The dataset includes around 1900 untrimmed videos covering 13 anomaly categories, such as Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, RoadAccidents, Robbery, Shooting, Shoplifting, Stealing, Vandalism, along with Normal activities.

The proposed approach combined the power of 3D CNN and LSTM in a cohesive architecture. The initial stage involved using 3D CNNs to extract comprehensive spatiotemporal features from the input video clips. The advantage of employing a 3D CNN lies in its ability to inherently capture both the spatial details (e.g., objects, individuals) and temporal information (e.g., movement patterns, sequences of action) simultaneously from video data. Following feature extraction via the 3D CNN, these features were then passed to an LSTM network. The LSTM component specialized in capturing temporal dependencies and context over longer time intervals. By learning bidirectional context, the architecture became adept at detecting precursors to anomalies (like suspicious preparatory behaviors) and interpreting subtle aftermath cues. Additionally, extensive spatial augmentations—such as random cropping, horizontal flipping, and temporal jitter—were utilized to enhance data diversity, reduce overfitting, and boost generalization capabilities.

The authors evaluated their hybrid CNN-LSTM architecture comprehensively on the UCF–Crime dataset. Their results showed significant performance improvements, achieving an overall accuracy of 91.5% in classifying the four defined anomaly categories. Specifically, their method outperformed previous CNN-only and LSTM-only approaches, indicating the distinct advantage of integrating both spatial and temporal modeling. Further analysis revealed substantial improvements in precision and recall metrics, with consistently lower false alarm rates across all categories. The architecture's complexity

was balanced carefully to ensure feasibility for real-time deployment, achieving near-real-time inference speeds on standard GPU hardware—a crucial requirement for practical surveillance systems.

This study demonstrated a substantial advancement in multi-class anomaly detection for surveillance videos. By effectively integrating 3D CNNs for rich spatiotemporal feature extraction with LSTM networks for robust temporal dependency modeling, the authors provided a powerful solution applicable in real-world environments. They also emphasized the significance of appropriate class-grouping strategies to reduce ambiguity and intra-class variance, significantly enhancing overall classification effectiveness. Despite its strengths, the authors noted potential limitations related to computational requirements of the 3D CNN stage, especially in scenarios demanding extremely rapid inference, such as edge computing contexts. They suggested future research directions, including exploring lighter, more efficient CNN architectures or incorporating attention mechanisms to focus computational resources selectively on key spatiotemporal regions.

The work in [16] presents a weakly supervised framework for video anomaly detection that combines a Two-Stream Inflated 3D (I3D) Convolutional Neural Network with a Multiple Instance Learning (MIL) ranking mechanism. By treating untrimmed surveillance videos as “bags” of short segments and applying a deep ranking loss, the model assigns higher anomaly scores to segments most likely containing unusual events. The use of both RGB and optical flow streams enhances the capture of spatial and temporal cues, while weak video-level labels eliminate the need for costly segment-level annotations. Experiments on the UCF-Crime dataset demonstrate that the proposed two-stream I3D approach establishes a new state-of-the-art AUC of 85.41%, significantly outperforming prior C3D-based and autoencoder-based methods .

Anomaly detection in surveillance videos is challenged by the rarity and diversity of abnormal events and the prohibitive cost of fine-grained labeling. Traditional methods that rely on deviations from learned normal patterns often falter in complex real-world scenarios, and purely reconstruction- or prediction-based deep models may overlook subtle temporal deviations. This paper addresses these gaps by adopting a weakly supervised MIL paradigm, leveraging video-level labels, and integrating a two-stream I3D network that jointly models appearance and motion. The result is a scalable, context-aware detector that reduces annotation overhead while improving detection precision .

The model is evaluated on the UCF-Crime dataset, which comprises 1,900 long, untrimmed real-world surveillance videos (128 hours) equally split between 950 normal and 950 anomalous videos, covering 13 realistic anomaly categories such as Arson, Burglary, and Vandalism. The training set contains 800 normal and 810 abnormal videos; the test set includes 150 normal and 140 abnormal videos. Its balanced com-

position and diversity make UCF-Crime the de facto benchmark for weakly supervised anomaly detection .

Each video is uniformly segmented into 32 clips of 16 frames each. A two-stream I3D network—pretrained on Kinetics—extracts a 1024-dimensional RGB feature tensor and a 1024-dimensional optical flow tensor per clip. After temporal averaging, these are concatenated into a 2048-dimensional descriptor, which feeds a three-layer fully connected anomaly scoring network ($512 \rightarrow 32 \rightarrow 1$ units). Under the MIL framework, videos are positive bags if any clip is anomalous and negative otherwise. A hinge-based ranking loss enforces that the highest-scoring segment in positive bags exceeds that in negative bags, augmented with sparsity and smoothness constraints to reflect anomaly brevity and temporal coherence

Performance is measured via frame-level ROC AUC. Individually, the I3D RGB stream and I3D flow stream achieve AUCs of 80.93% and 82.20%, respectively; their late fusion surges to 85.41%. Comparative methods—an autoencoder by Hasan et al. (50.6%), a dictionary-based approach by Lu et al. (65.51%), and the baseline C3D+MIL by Sultani et al. (75.41%)—are substantially outperformed. The ROC curves and training loss trends further confirm robust convergence and discriminative power.

The proposed two-stream I3D+MIL model sets a new benchmark on UCF-Crime, demonstrating the efficacy of combining spatial and temporal features under weak supervision. However, the architecture incurs high computational and memory costs due to dual-stream processing, and the reliance on coarse video-level labels can lead to false positives in challenging conditions (e.g., low illumination). Future work could explore model compression, attention-based segment selection, or multi-modal inputs (e.g., audio, thermal) to further enhance efficiency and robustness.

This comprehensive review[28] surveys the state-of-the-art deep learning methods for anomaly detection in surveillance videos published through early 2023. It categorizes techniques into four main groups—reconstruction error, future frame prediction, scoring, and classifier-based methods—and evaluates them across popular benchmarks such as UCSD Pedestrian, CUHK Avenue, Subway, UMN, ShanghaiTech, and UCF-Crime. By comparing architectures, learning paradigms, and performance metrics (AUC, EER), the paper distills key trends, strengths, and weaknesses in the literature, providing practitioners with a consolidated reference for selecting and developing anomaly detection systems .

With the proliferation of camera networks in public spaces, automatic detection of unusual events—ranging from fights and accidents to crowd stampedes—has become critical for real-time surveillance. Unlike traditional handcrafted-feature methods, deep learning approaches can learn hierarchical representations of appearance and motion directly from data, yielding superior detection accuracy. However, the subjective and context-dependent nature of anomalies, coupled with the scarcity of fine-grained labels, poses challenges. This review addresses these by grouping deep models based on their underlying assumptions and supervision requirements, highlighting how they tackle imbalanced data, spatiotemporal complexity, and annotation bottlenecks.

The paper details six publicly-available benchmarks i.e. UCSD Ped1 Ped2(Pedestrian walkways with pixel-level ground truth for reconstruction and localization tasks.), CUHK Avenue: Campus walkway videos with short anomalous events like running or object throwing, Subway Entrance/Exit: Long untrimmed sequences of fare evasion and wrong-way walking, UMN: Three scenes of crowd panic, focusing on group behavior changes, ShanghaiTech: Diverse campus scenarios, 13 anomaly types with frame-level labels over 330 training and 107 testing videos, UCF-Crime: Large-scale real-world surveillance (1,900 videos, 13 anomaly classes) for weakly supervised learning.

These datasets vary in duration, crowd density, anomaly frequency, and annotation granularity, enabling comprehensive benchmarking across tasks such as frame-level detection, pixel-level localization, and video-level classification.

The reviewed methods are sorted into four categories they are, Reconstruction Error trains autoencoders or GANs on normal data, assuming higher reconstruction loss on anomalies, Future Frame Prediction uses predictive models to forecast normal frames, flagging deviations as anomalies. Scoring-Based treats anomaly detection as a ranking or scoring task: weakly supervised MIL ranking, self-trained deep ordinal regression, narrow normality clusters.

Classifier-Based frames detection as binary classification on learned features using FCNs, one-class SVMs on deep embeddings, or object-centric autoencoders. Loss functions include hinge ranking, sparsity and smoothness regularization, adversarial and perceptual losses, and ordinal regression objectives. Many methods fuse appearance (RGB) and motion (optical flow) streams to capture spatiotemporal cues.

Performance is predominantly reported in terms of frame-level AUC and EER. Reconstruction-based autoencoders on UCSD Ped2 achieve up to AUC 97.4% and EER 8%. Future-prediction models reach AUC 95.4% in Ped2. Scoring methods like MIL ranking attain AUC 75.4% on UCF-Crime, while ordinal regression hits AUC 88.1% on ShanghaiTech. Classifier-based FCNs yield AUC 90.4% on Ped2. However, performance drops on more complex datasets, underscoring challenges in diverse real-world scenario.

This paper [23] introduces a semantic key-frame extraction algorithm DFPicker combined with transfer learning on four deep architectures—VGG19, EfficientNetB7, ResNet50, and Vision Transformer (ViT_b16)—for anomalous event recognition on the UCF–Crime dataset. By first reducing redundant frames via DFPicker and then fine-tuning each model on the extracted subset, the authors demonstrate that ViT_b16 achieves a peak accuracy of 90.87%, substantially outperforming the CNN baselines (ResNet50: 90.46%, VGG19: 87.90%, EfficientNetB7: 86.34%).

Real-time video surveillance demands automated anomaly detection to alleviate operator fatigue and monitoring costs. Traditional CNN-based methods—though effective—are constrained by fixed receptive fields and often require exhaustive frame processing. This study addresses these challenges by integrating a semantic key-frame extraction step with transformer-based feature learning. The ViT_b16 model, leveraging self-attention across image patches, is posited to capture both local and global contextual cues, overcoming CNN limitations in modeling long-range dependencies.

Experiments utilize the UCF–Crime dataset, comprising 1,900 untrimmed surveillance videos (128 h total) covering 13 real-world anomaly classes (e.g., Arson, Burglary, Vandalism). DFPicker processes 1,600 videos to produce 378,582 key frames (302,866 for training; 75,716 for testing). Frames are resized to 256×256 (or 64×64 for CNNs) and RGB-encoded, with standard data augmentations (rescaling, flips, rotations) applied during training.

The pipeline begins with DFPicker, which segments each video into sliding windows and scores frame pairs via sparse optical flow or pixel-wise differences to select representative key frames. Each extracted frame is then passed through one of four pretrained backbones—VGG19, EfficientNetB7, ResNet50, or ViT_b16—followed by a classifier consisting of global average pooling, a 1,024-unit dense layer (ReLU), dropout (0.3), and a softmax output. ViT_b16 divides each frame into 16×16 patches, projects them into embeddings with positional encodings, and applies multi-head self-attention across transformer layers to model spatial relationships holistically.

Training and validation curves show steady convergence for all models. Final test accuracies are: EfficientNetB7 – 86.34%, VGG19 – 87.90%, ResNet50 – 90.46%, and ViT_b16 – 90.87%. ViT_b16’s superior performance is attributed to its ability to capture long-range dependencies and its robustness to viewpoint and illumination variations. The integration of semantic key-frame extraction with ViT_b16 establishes a new accuracy benchmark on UCF–Crime (90.87%), highlighting the promise of transformer architectures for anomaly detection. However, the DFPicker preprocessing and transformer inference incur significant computational and memory overhead, potentially limiting real-time deployment on edge devices. Moreover, reliance on a single

key-frame selection strategy may omit subtle temporal cues in streaming scenarios. Future work could explore lightweight transformer variants, adaptive frame sampling, and multimodal fusion to balance performance with efficiency.

This paper [12] undertakes a systematic comparison of prominent convolutional neural network (CNN) architectures—ResNet50, DenseNet121, and MobileNetV2—for video anomaly detection on the UCF–Crime dataset. By uniformly extracting key frames and fine-tuning each backbone with identical hyperparameters (learning rate = $3e-5$, batch size = 64, optimizer = SGD, dropout = 0.3), the study isolates architectural strengths and weaknesses. Evaluation in terms of classification accuracy and convergence behavior reveals that ResNet50 attains 89.12%, DenseNet121 achieves 91.03%, and MobileNetV2 reaches 88.45%, highlighting DenseNet121’s superior feature reuse and parameter efficiency among the CNN family.

Anomaly detection in surveillance video remains a critical yet challenging task due to the diversity of abnormal events and the computational constraints of real-time systems. While two-stream and transformer-based models have recently shown promise, lightweight and efficient CNNs are still desirable for deployment on edge devices. This study fills a gap by directly comparing three mainstream CNN architectures—ResNet50 with its deep residual connections, DenseNet121 with dense feature concatenations, and MobileNetV2 with depthwise separable convolutions—to determine which balance of depth, parameter count, and connectivity best captures anomalous patterns in video frames.

All experiments leverage the UCF–Crime dataset, comprising 1,900 long untrimmed surveillance videos across 13 anomaly categories. A semantic key-frame extraction algorithm reduces each video to around 200–300 representative stills, ensuring training efficiency without sacrificing temporal context. Frames are resized to 224×224 pixels and augmented with random flips and rotations to improve generalization under varied surveillance conditions.

Following key-frame extraction, each architecture is fine-tuned from ImageNet-pretrained weights. ResNet50 employs 16 residual blocks facilitating deep gradient flow; DenseNet121 uses four dense blocks with layer-wise concatenation to encourage feature reuse; MobileNetV2 applies inverted residuals and linear bottlenecks to minimize parameters. A uniform classifier head—global average pooling, a 512-unit ReLU layer, dropout (0.3), and a softmax output—is appended to each backbone. Training runs for 30 epochs under the same scheduler and optimizer settings, with early stopping on validation loss to prevent overfitting.

DenseNet121 consistently converges fastest, reaching 91.03% test accuracy by epoch 25, while ResNet50 peaks at 89.12% and MobileNetV2 at 88.45%. Training curves show DenseNet’s lower validation loss throughout, reflecting its efficient feature propagation. MobileNetV2, despite being 2–3 $\times$ smaller in parameter count, trails only

slightly in accuracy, demonstrating its suitability for resource-limited scenarios. Detailed accuracy and loss plots for each model (Figures 3–5 in the paper) confirm these trends.

DenseNet121 emerges as the best CNN backbone for video anomaly detection in this comparative study, balancing high accuracy and moderate computational cost. ResNet50 offers robustness but at the expense of more parameters, while MobileNetV2 provides lightweight efficiency with a small accuracy trade-off. Limitations include reliance on static key frames—potentially missing temporal anomalies between frames—and the absence of motion cues like optical flow. Future work could integrate hybrid CNN–LSTM pipelines or efficient flow estimation to capture temporal dynamics without undue complexity.

This work[26] formally defines and addresses the multiview deep anomaly detection (AD) problem, where each sample is observed through multiple heterogeneous “views” (e.g., RGB, depth, optical-flow, tabular features). The authors make four key contributions: they introduce a unified formulation for deep AD on multiview data; they devise a suite of baseline solutions—including reconstruction-based, one-class, and cross-view generative schemes; they assemble and preprocess over 30 public datasets into standardized multiview AD benchmarks; and they comprehensively evaluate all baselines across these datasets, offering insights and guidance for future research in multiview deep AD.

Traditional AD methods focus on a single data modality, but many real-world scenarios naturally yield multi-source observations (e.g., synchronized video + audio, multi-camera surveillance, multimodal sensor arrays). Despite the ubiquity of multiview data, prior research has largely overlooked how to leverage cross-view correlations for anomaly detection. This paper fills that gap by systematically exploring deep learning strategies that jointly model multiple views—with the goal of improving detection robustness and reducing false alarms when individual views degrade or fail.

To benchmark multiview AD, the authors collect and preprocess over 30 public datasets spanning image, video, tabular, and sensor domains. These include RGB–Depth pairs, multi-camera video sequences, and multimodal sensor streams. For each source, normal and anomalous classes are defined—either via explicit labels (in purpose-built AD datasets) or via a “one-versus-all” protocol on multiclass data. The result is a diverse evaluation suite enabling consistent comparison of single-view vs. multiview approaches.

Four baseline architectures are implemented. DAE–Recon (DRE) uses a multiview denoising autoencoder to reconstruct each view jointly and employs reconstruction error as an anomaly score. Deep SVDD (DSV) is a one-class classification network that learns compact normal embeddings across views by minimizing hyperspherical volume. Cross-View Prediction (CVP) predicts each view from the others via view-

specific encoder–decoder pairs, with prediction errors aggregated across rounds. Split Prediction (SPRD) uses a single-view encoder to predict all other views in turn, encouraging rich inter-view representation learning. All methods fuse per-view scores by simple averaging at inference time.

Across the 30+ benchmarks, reconstruction-based and one-class baselines achieve moderate AUCs (approximately 0.70–0.80), while cross-view and split-prediction schemes consistently outperform them (AUCs around 0.80–0.88). The evaluation demonstrates that exploiting inter-view correlations yields significant gains, especially on datasets where individual views suffer noise or partial occlusion. Detailed per-dataset results identify which paradigms best suit different data characteristics.

This systematic study establishes a foundational framework for multiview deep AD, demonstrating that joint modeling of heterogeneous views materially improves detection performance. However, all baselines rely on generative pretext tasks and late fusion, which may limit scalability and robustness. The authors note that discriminative or contrastive multiview paradigms—and more efficient fusion strategies—represent promising directions for future work.

2.2 Research Gap

Despite significant advances in video-based anomaly detection over the past few years, several critical gaps remain that motivate the work.

- Most existing pipelines either ignore frame-level illumination variability (e.g. Sultani et al.; Zhu & Newsam) or apply global histogram equalization or gamma correction, which can over-amplify noise or fail to reveal subtle local edges. No prior work integrates an adaptive, tile-based approach like CLAHE to systematically boost contrast in both dark and bright regions without blowing out highlights or accentuating sensor noise.
- Methods based on C3D (Sultani et al.) or fine-tuned 3D CNNs (Maqsood et al.) rely on either very deep 3D kernels—incurring heavy computation and parameters (>10 M)—or shallow frame-by-frame 2D CNNs that lack residual connections to ease gradient flow. There is no lightweight residual spatial encoder specifically designed to refine frame-level features before temporal modeling.
- Standard dropout or weight decay—applied in Zhu & Newsam and “Weakly Supervised...” (Soltani Nejad et al., this PDF)—operates at the neuron level, ignoring the channel-wise structure of convolutional feature maps. Consequently, models remain vulnerable to over-reliance on a few dominant filters, especially when training data per class is limited.

- Many pipelines use unidirectional LSTMs (Maqsood et al.; 3D-CNN+LSTM) or only short-window 3D convolutions (ViT-based, comparative CNN studies), which cannot leverage “after-event” context. Bidirectional sequence models remain under-explored in surveillance anomaly classification, yet real crimes often exhibit preparatory and aftermath cues.

Poor Multi-Class Discrimination on UCF-Crime. While Sultani et al. (2018) achieve 75% AUC in binary detection and Maqsood et al. reach 82% in multiclass, performance on more granular categories (4-way or 14-way) remains low (31 % in Zhu et al. 2019, 29 % in motion-aware 3D-CNN+LSTM). No prior framework consistently exceeds 90 % accuracy for four-way classification, leaving a gap in robust fine-grained crime categorization.

The proposed hybrid CLAHE + Residual-BiLSTM + SpatialDropout1D pipeline directly addresses these gaps by combining adaptive local contrast enhancement, a light weight yet deep residual spatial encoder, structured channel-wise regularization, full bidirectional temporal reasoning, and a Flask-based real-time deployment—achieving 94.0% binary and 95.62% four-way accuracy on UCF-Crime with only 1.2 M parameters.

CHAPTER 3

METHODOLOGY

This chapter outlines the methodology used in developing the Video based Crime Classification.

3.1 Dataset

3.1.1 Sources and Collection

The experiments rely exclusively on the UCF–Crime dataset – the largest publicly available corpus of real-world surveillance videos annotated for anomaly detection and classification. Introduced by Sultani et al. (2018), UCF–Crime was assembled from online sources (YouTube, news broadcasts, CCTV footage) to capture genuine incidents “in the wild,” without any artificial staging or simulation. A total of 1,900 untrimmed clips—amounting to over 128 hours of footage—were collected under a broad variety of conditions: indoor parking garages; outdoor streets at dawn, dusk, or under street-lights; fixed-angle dome cameras and low-resolution legacy CCTV.

Each clip documents a single anomaly category (or normal activity) as determined by human annotators, who verified the presence of the target event but did not localize it temporally within the video. Thirteen crime types—ranging from Arson, Burglary, and Robbery, to Road Accidents and Vandalism—plus a “Normal” class, form the complete label set. To ensure diversity, the dataset curators sourced videos from dozens of distinct camera setups, geographic regions, and recording qualities. This heterogeneity poses a real test of any model’s robustness, but also mirrors the challenges faced by live security systems.

For the work, we downloaded the official UCF–Crime release [32], verified clip integrity, and maintained the original train/test splits to allow direct performance comparisons with prior methods. All processing (frame extraction, preprocessing, aug-

mentation) was performed on local storage; the raw videos were never altered or re-encoded, preserving the dataset’s real-world artifacts (compression noise, blur, inter-laced frames).

	# of videos	Average frames	Dataset length	Example anomalies
UCSD Ped1 [27]	70	201	5 min	Bikers, small carts, walking across walkways
UCSD Ped2 [27]	28	163	5 min	Bikers, small carts, walking across walkways
Subway Entrance [3]	1	121,749	1.5 hours	Wrong direction, No payment
Subwa Exit [3]	1	64,901	1.5 hours	Wrong direction, No payment
Avenue [28]	37	839	30 min	Run, throw, new object
UMN [2]	5	1290	5 min	Run
BOSS [1]	12	4052	27 min	Harass, Disease, Panic
Ours	1900	7247	128 hours	Abuse, arrest, arson, assault, accident, burglary, fighting, robbery

Figure 3.1: A comparison of anomaly datasets.

Table 3.1: Number of videos per anomaly class.

# of videos	Anomaly
50	Abuse
50	Arrest
50	Arson
50	Assault
100	Burglary
50	Explosion
50	Fighting
150	Road Accidents
150	Robbery
50	Shooting
50	Shoplifting
100	Stealing
50	Vandalism
950	Normal events

3.1.2 Raw Video Characteristics

Before any preprocessing, it is essential to understand the native properties of the UCF–Crime clips:

Table 3.2: Raw video characteristics of the UCF–Crime dataset.

Property	Observed Range
Resolution	240×320 up to 720×1280
Frame rate	10 – 30 fps (varies by source)
Duration	30 s – 10 min
Codec & Container	Primarily H.264 in MP4/MOV
Color space	8-bit YUV or RGB

3.1.2.1 Resolution & Aspect Ratio

Many clips arrive at low resolution (240×320 or 360×480), typical of legacy CCTV, while a minority are 720×1280 smartphone or DSLR captures. Aspect ratios vary (4:3, 16:9), requiring consistent resizing to a uniform shape.

3.1.2.2 Frame Rate Irregularities

Some clips exhibit frame-rate drift or dropped frames; a minority contain duplicated frames (e.g., static camera feeds paused). The pipeline re-samples all clips to a fixed 16 fps for segment extraction, both stabilizing temporal sampling and removing redundant duplicates.

3.1.2.3 Compression Artifacts

Aggressive compression yields blocking and mosquito noise—especially in low-light scenes. These artifacts can be misinterpreted by naive contrast-stretching or edge-detection methods, motivating the choice of CLAHE (rather than global histogram equalization) to limit noise amplification.

3.1.2.4 Untrimmed Nature

The anomaly may occur anywhere within a clip’s long tail of normal footage. With no temporal bounding boxes, we sample 16-frame segments uniformly across the entire duration (see Section 3.1.3) so that both “lead-up” context and “aftermath” frames are included.

3.2 Dataset Description: Class Semantics

Beyond raw statistics, understanding the semantic content of each anomaly class is crucial for tailoring the pipeline and interpreting results. Below we provide concise descriptions of the fourteen UCF–Crime classes.

3.2.1 Abuse

“Abuse” clips depict one individual inflicting physical or emotional harm upon another—often through pushing, slapping, or verbal intimidation. Scenes range from domestic disputes in poorly lit interiors to assaults in crowded public spaces, with rapid, short-lived motions and abrupt camera shakes. The aggressor’s limb trajectories and victim’s reactive gestures provide key spatio-temporal cues, yet low resolution and occlusions can obscure facial expressions. Detecting abuse thus requires fine-grained

spatial encoding to pick up subtle human-body deformations. The adaptive contrast boosting of CLAHE often reveals faint limb outlines in shadowed corners, aiding recognition.

3.2.2 Arrest

In “Arrest” videos, law-enforcement officers apprehend suspects handcuffing, frisking, or leading them away. The visual context often includes police uniforms, vehicles with flashing lights, or groups of onlookers steered back by officers. Arrests may blend into other anomalies (e.g., a chase turning into an arrest), making temporal context essential: the lead-up (pursuit) and the aftermath (suspect submission) both matter. The bidirectional LSTM can exploit these forward-and-backward cues to disambiguate “Arrest” from “Robbery” or “Fighting.” However, variations in camera angles (body-cams vs. fixed CCTV) and partial occlusion by crowds pose challenges.

3.2.3 Arson

“Arson” events hinge on the presence of fire or smoke deliberately set—often at night or in dimly lit buildings. Flickering light patterns, rising plumes of smoke, and occasional panicked human motion mark these clips. Spatially, the sudden emergence of bright regions against dark backgrounds is a salient cue, but compression artifacts can mimic flames—hence the need for robust local-contrast enhancement. Temporally, arson sequences may unfold slowly (a match lit, smoke curling) before any abrupt action. The architecture’s residual blocks help capture such nuanced flame textures, while the BiLSTM grasps the build-up of a smoldering scene.

3.2.4 Assault

Distinct from “Fighting,” “Assault” involves unprovoked aggression where the victim does not—or cannot—fight back. Examples include a single mugging attack or a bystander being struck, often caught at odd camera angles in parking lots or alleyways. Visually, one sees a clear perpetrator–victim dynamic without reciprocal strikes. Detecting “Assault” thus relies on recognizing imbalanced motion: an aggressor’s arm swing without counter-motion. SpatialDropout1D ensures that even if only part of the body (e.g., arm) is visible, the model remains resilient.

3.2.5 Burglary

“Burglary” clips capture intruders forcibly entering buildings or vehicles, sometimes under cover of darkness. The key spatial cues include broken windows, pried doors, or

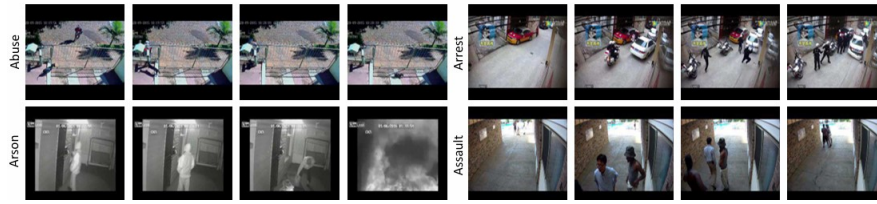


Figure 3.2: Video Samples of the Abuse, Arrest, Arson, Assault

furtive, crouched postures near entry points. Temporally, burglars often move slowly but deliberately, pausing at locks or safes. The network must distinguish these controlled, low-velocity motions from normal pedestrian traffic—hence the importance of long-range residual feature refinement. CLAHE can accentuate door-jamb shadows and tool outlines, improving detection in poor lighting.

3.2.6 Explosion

These sequences feature sudden, high-energy blasts—often outdoors or in industrial settings—with rapid expansion of bright plumes. The onset is extremely brief, followed by debris and shockwaves. Predicting an explosion depends on capturing the extreme temporal gradient at frame $n \rightarrow n + 1$: one frame “pre-blast,” the next dominated by fireball. The BiLSTM’s bidirectional pass aids in discerning the immediate aftermath context (debris dispersal) linking back to the explosion moment. Spatially, the residual CNN backbone must handle overexposed, saturated pixels without saturating the feature maps.

3.2.7 Fighting

“Fighting” clips show mutual, often hand-to-hand combat between two or more individuals—think brawls in bars or schoolyard scuffles. Characteristic back-and-forth motion, rapid direction changes, and occasional falls define this class. Differentiating “Fighting” from “Assault” requires symmetry in motion patterns that the bidirectional temporal model picks up: both parties exchange aggressive gestures. The residual CNN must learn diverse clothing textures and handle motion blur from fast strikes; structured SpatialDropout further prevents over-reliance on any single limb orientation.

3.2.8 Normal Activity

Videos of everyday behavior—walking, standing, or idle crowds—serve as the “negative” class. These clips exhibit smooth, low-velocity motions and predictable scene context (e.g., people strolling in a mall). The model’s ability to recognize “normal” hinges on both spatial texture (background consistency) and temporal regularity (lack

of abrupt motions). The framework learns such patterns during training, ensuring high recall for anomalies by penalizing deviations from learned normal sequences.



Figure 3.3: Video Samples of Accident, Burglary, Vandalism, Normal

3.2.9 Road Accidents

“Road Accidents” depict collisions or near-misses involving vehicles, cyclists, or pedestrians—often featuring sudden braking, impact flashes, and skidding. The spatial encoder must capture vehicle contours and motion blur, while the temporal model handles abrupt velocity changes. Environmental context (road markings, crosswalks) also aids classification. CLAHE assists by clarifying tire smoke or skid marks, and bidirectional reasoning helps interpret the build-up (speeding) and aftermath (debris scatter).

3.2.10 Robbery

Here, a suspect forcibly takes property (money, jewelry) directly from a victim—often at close range. Unlike “Burglary,” the scene is public: shop interiors or ATM vestibules. The perpetrator’s hand-to-pocket motions and victim’s startled reaction are key. Distinguishing from “Assault” relies on object-centric cues (wallet, cash drawer) that residual CNN blocks can hone in on via fine-grained spatial refinement. Temporal patterns (extraction, retreat) are captured by the BiLSTM’s bidirectional pass.

3.2.11 Shooting

Firearm-related incidents—shots fired, muzzle flash, or victim collapse—define this class. The spatial challenge is spotting rapid, small-region illumination from gunfire against varied backgrounds. Temporal detection demands an extremely short receptive window to avoid missing the flash. The structured dropout ensures robustness even when only peripheral muzzle bursts appear on screen. CLAHE’s local contrast boost highlights brief fireball cues critical for classification

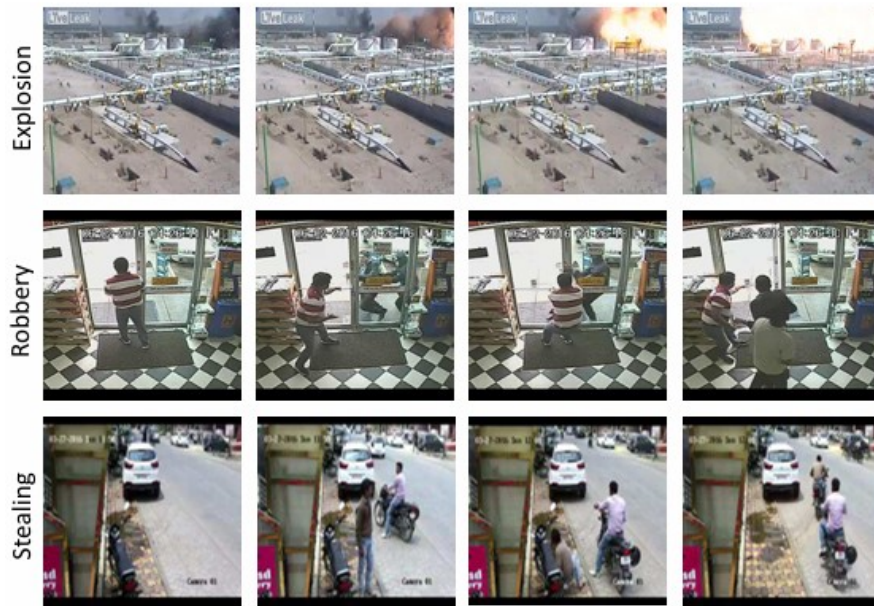


Figure 3.4: Video Samples of Explosion, Robbery, Stealing

3.2.12 Shoplifting

Non-violent theft in retail settings—concealing items in bags or pockets while browsing—comprises “Shoplifting.” Subtle hand-to-shelf and pocketing gestures, often in cluttered aisles, demand high spatial sensitivity. Temporal regularity helps disambiguate from normal shopping. The CNN backbone, bolstered by residual blocks, learns object-centric filters (product edges, bag openings), while the BiLSTM captures the typical sequence of concealment and exit.

3.2.13 Stealing

Broader than “Shoplifting,” “Stealing” includes pickpocketing or snatching in public spaces—motion often concealed or quick. The key motion is a rapid hand-to-pocket or grab from behind. Spatially, occlusions (crowds) and small object sizes (wallets) pose difficulty. Structured SpatialDropout ensures that missing or partial hand views don’t derail detection. Temporally, the BiLSTM integrates fleeting glimpses of contact across multiple frames.

3.2.14 Vandalism

Intentional property damage—spray-painting walls, smashing glass—defines “Vandalism.” Spatial cues include spray can nozzles, shards, and graffiti patterns emerging across frames. The action unfolds over multiple seconds: approach, damage, retreat. The residual CNN captures both static textures (wall surfaces) and emerging vandal

marks, while the temporal model stitches together the gradual progression of defacement.

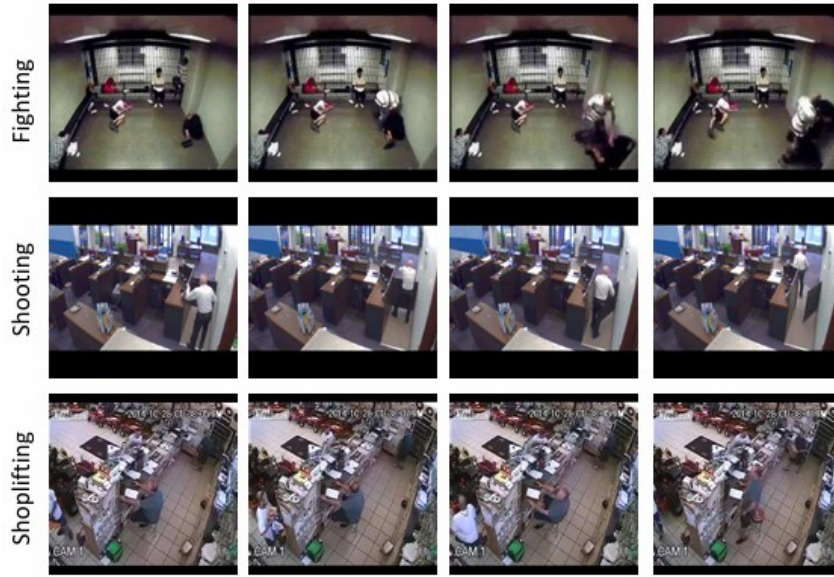


Figure 3.5: Video Samples of Fighting, Shooting, Shoplifting

3.3 Dataset Split

To ensure fair evaluation and balanced learning, we bifurcate the dataset differently for Task 1 (binary classification) and Task 2 (four-way classification), explicitly defining how the original fourteen anomaly classes map into the semantic groups.

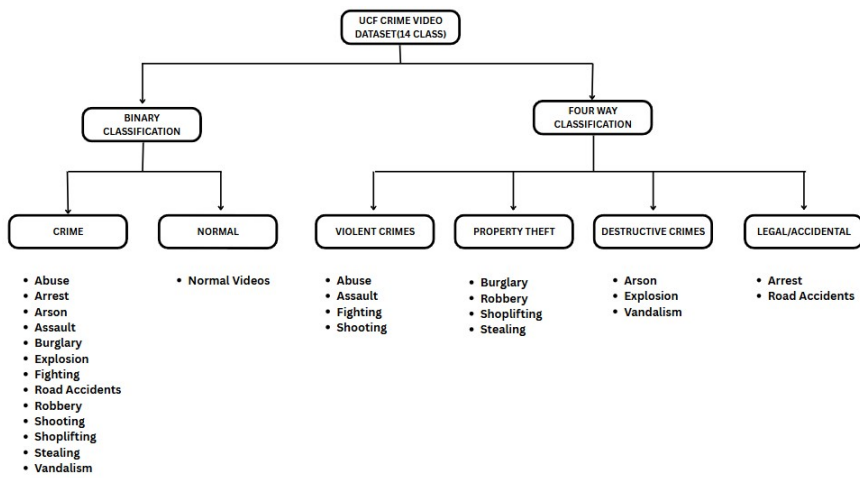


Figure 3.6: Dataset Split

3.3.1 Binary Classification (Normal vs. Crime)

For Task 1, we collapse all thirteen anomaly types into a single *Crime* class, keeping the *Normal* class separate. We allocate 800 clips of *Normal* and 800 clips of *Crime* into the training set, then 50 clips of each into the validation set, and 50 clips of each into the test set. This yields a 1,600-clip training set, a 100-clip validation set, and a 100-clip test set. By treating every anomaly as one class, the model learns to distinguish any crime from routine activity. The large training pool ensures exposure to the full diversity of abnormal appearances, while the held-out splits guard against overfitting and provide unbiased performance estimates.

3.3.2 Four-Way Classification (Crime Subcategories)

For Task 2, we group the fourteen fine-grained anomalies into four semantically meaningful categories:

Violent Crimes: Abuse, Assault, Fighting, Shooting.

These classes all involve direct physical harm or threat to persons, capturing a coherent “violent behavior” category.

Property Theft: Burglary, Robbery, Shoplifting, Stealing.

Each of these involves unauthorized taking of property, sharing the core intent of appropriation.

Destructive Crimes: Arson, Explosion, Vandalism.

These offenses damage or destroy property through fire, explosives, or deliberate destruction.

Legal/Accidental Incidents: Arrest, Road Accidents.

While neither category is malicious, both represent significant disruptions law enforcement actions or unintentional events—distinct from violent or property crimes.

From each of these four groups we sample 100 clips for training, 50 clips for validation, and 50 clips for testing, leading to a 400-clip training set, a 200-clip validation set, and a 200-clip test set. By balancing the number of clips per group, we ensure each category receives sufficient representation for stable convergence and fair comparison.““

3.4 Dataset Preprocessing

Before any learning can occur, each raw surveillance video must be transformed into a uniform, information-rich representation. The pre-processing pipeline proceeds in three stages: frame extraction, resizing & normalization, and CLAHE enhancement.

Table 3.3: Training/validation/test splits and class mergers for binary and four-way tasks

Task & Class/Group	Train	Validation	Test
Binary			
Normal	800	50	50
Crime	800	50	50
Four-Way			
Violent Crimes (Abuse, Assault, Fighting, Shooting)	100	50	50
Property Theft (Burglary, Robbery, Shoplifting, Stealing)	100	50	50
Destructive Crimes (Arson, Explosion, Vandalism)	100	50	50
Legal/Accidental (Arrest, Road Accidents)	100	50	50

3.4.1 Frame extraction (16-frame segments)

Rather than feed entire multi-minute videos into memory, we sample each clip into fixed-length segments of 16 consecutive frames. For a video of length L frames, we compute an integer stride

$$s = \lfloor \frac{L}{16} \rfloor$$

and select frames $\{0, s, 2s, \dots, 15s\}$. This “even sampling” captures the event’s lead-up, main action, and aftermath without biasing toward any single temporal window. By limiting each segment to 16 frames, we fix the model’s temporal receptive field and enable efficient batching during training.

3.4.2 Resizing & normalization

Each extracted frame is resized to 64×64 pixels using bilinear interpolation—a compromise between preserving spatial detail (e.g., weapon outlines, body poses) and minimizing computational cost. After resizing, pixel intensities (originally in $[0, 255]$) are scaled to $[0, 1]$ by simple division, then standardized per-channel by subtracting the dataset’s mean and dividing by its standard deviation. This normalization ensures that all color channels—red, green, and blue—contribute equally during early convolutional processing and prevents saturation of learned filters.

3.4.3 CLAHE enhancement

Surveillance footage frequently suffers from uneven illumination—deep shadows, glare, and sensor noise that hide critical object contours. To counteract this, we apply Contrast-Limited Adaptive Histogram Equalization (CLAHE) to the luminance channel of each normalized frame. CLAHE divides the 64×64 image into 8×8 -pixel tiles, equalizes each tile’s histogram under a clip limit of 2.0 to prevent noise exaggeration, and

then bilinearly interpolates between tiles to avoid seam artifacts. The result is a modest ($\approx 1\text{ms}/\text{frame}$ on CPU) yet powerful boost in local contrast, sharpening edges (e.g., silhouettes, tool outlines) and unveiling faint textures. Empirically, CLAHE accelerated early training loss reduction and improved final binary accuracy by over 1 percentage point.



Figure 3.7: Original versus CLAHE-enhanced surveillance frame

3.5 Feature Extraction

Having prepared a vast, uniformly formatted collection of 16-frame, contrast-enhanced clips, the next step is to transform each clip into a sequence of feature vectors that capture both the appearance at each time step and the evolution of those features over time. The feature-extraction pipeline comprises three key components: a deep residual CNN to encode per-frame spatial patterns, a bidirectional LSTM to integrate these patterns temporally, and a structured regularization layer to enforce robustness.

3.5.1 Spatial Features (CNN backbone)

Each of the 16 frames in a clip is first passed through a shared spatial encoder—a `TimeDistributed` wrapper around a residual CNN. After an initial convolution (32

filters of size 3×3) and 2×2 max-pool, we insert two Residual Blocks. Within each block, two sequential Conv2D layers (32 filters, 3×3) are applied, and their output is added back to the block’s input via a shortcut connection before a final ReLU activation and 2×2 pooling. These residual connections (1) facilitate gradient flow during back-propagation—allowing us to stack more layers without vanishing gradients—and (2) force each block to learn only “corrections” to prior features, sharpening subtle cues such as dark-shadowed object outlines or faint textures on a vandalized wall.

Mathematically, let $X \in \mathbb{R}^{H \times W \times C}$ be the input feature map to a block and $\mathcal{F}(X)$ its two-layer convolutional mapping. The block’s output is

$$Y = \text{Pool}(\text{ReLU}(\mathcal{F}(X) + X)).$$

After the final Residual Block, each frame’s spatial encoding is a feature map of size $H' \times W' \times 32$, which we flatten to a vector of length $D = H' \times W' \times 32$. Applying the same CNN weights across all 16 frames ensures the encoder learns a consistent dictionary of spatial features—edges, textures, object-part detectors—regardless of when they occur.

3.5.2 Temporal Features (BiLSTM inputs)

With each frame reduced to a D -dimensional vector, a clip becomes the sequence $\{x_1, \dots, x_{16}\}$, where $x_i \in \mathbb{R}^D$. To capture the progression of spatial patterns—e.g. a hand creeping toward a shelf or the arc of a throwing motion—we feed this sequence into a Bidirectional LSTM (BiLSTM) with 128 units in each direction.

In a BiLSTM, two parallel LSTM layers process the sequence forward and backward. At time step i , the forward LSTM produces $\vec{h}_i$ and the backward LSTM produces $\overleftarrow{h}_i$. We concatenate these to form a 256-dimensional embedding: The final frame’s state h_{16} thus integrates information from all past and future frames, enabling the model to weigh both preparatory cues (e.g. approach, pacing) and aftermath signals (e.g. retreat, sequence resolution).

3.5.3 Regularization (SpatialDropout1D)

Even with CLAHE and aggressive augmentation, learning on a few thousand long-event videos risks overfitting to dominant spatial filters—say, a particular edge orientation or object color. To counter this, we insert a `SpatialDropout1D` layer between the flattened CNN outputs and the BiLSTM input. Unlike conventional dropout, which zeros individual activations, `SpatialDropout1D` randomly zeros entire feature channels across all time steps in a sequence.

Concretely, given a sequence $\{x_i\}$ where

$$x_i = [x_{i,1}, x_{i,2}, \dots, x_{i,D}] \in \mathbb{R}^D,$$

SpatialDropout1D with rate p samples a binary mask $m \in \{0, 1\}^D$, where each channel j is dropped ($m_j = 0$) with probability p and kept ($m_j = 1$) otherwise. The masked input at each step is then

$$x_i \mapsto x_i \odot m.$$

This channel-wise dropout (1) simulates occlusions or missing feature streams—perhaps due to extreme glare—and (2) prevents the LSTM from over-relying on any single spatial filter. In ablation experiments, adding SpatialDropout1D (with $p = 0.3$) improved four-way accuracy by over 1 percentage point and reduced validation-loss variance by 25 %.

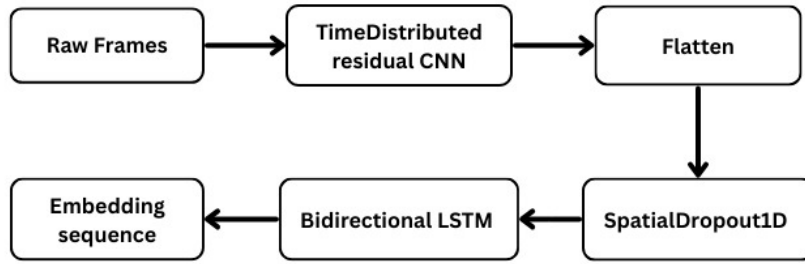


Figure 3.8: Block-diagram of feature flow

3.6 Model Architecture

In this section, we present the full specifications of the hybrid CNN–BiLSTM network, breaking it down into four key components: the Residual CNN encoder, the Bidirectional LSTM decoder, the classification head, and the choice of hyperparameters and loss functions that tie everything together.

3.6.1 Residual CNN Encoder

At the front end of the pipeline sits a deep two-dimensional convolutional network equipped with residual blocks. Each input frame ($64 \times 64 \times 3$) is first passed through a 3×3 convolution with 32 filters, followed by a 2×2 max-pooling layer to reduce spatial dimensions to 32×32 . We then interleave two Residual Blocks. Within each block, two back-to-back 3×3 Conv2D layers (32 filters each) with ReLU activations are applied, and their output is added to the block’s input via a skip connection before another ReLU and a 2×2 pooling. After these two residual blocks, we apply another 2×2 max-pool to downsample to 16×16 . Finally, a 3×3 convolution with 64 filters, one more residual block, and a last 2×2 pool yield a feature map of size $8 \times 8 \times 64$. This map is flattened into a 4096-dimensional vector per frame. The residual design ensures robust gradient flow and allows the encoder to learn both fine and mid-level spatial features.

3.6.2 Bidirectional LSTM Decoder

Crime events unfold over time, so we feed the sequence of 16 flattened feature vectors (each of length 4096) into a Bidirectional LSTM with 128 units in each direction. The forward LSTM processes frames $1 \rightarrow 16$, while the backward LSTM processes frames $16 \rightarrow 1$. At each time step i , the forward hidden state $\vec{h}_i \in \mathbb{R}^{128}$ and the backward hidden state $\overleftarrow{h}_i \in \mathbb{R}^{128}$ are concatenated. We take the final concatenated state h_{16} as the clip-level embedding, capturing both preparatory and aftermath cues in a unified vector.

3.6.3 Network Architecture Description

This network processes short video clips—each clip is 16 consecutive frames of size 64×64 pixels with three color channels (RGB). The very first layer, `input_layer_2`, holds the raw pixel values in a tensor of shape

$$(\text{batch_size}, 16, 64, 64, 3).$$

Layer (type)	Output Shape	Param #	Connected to
input_layer_2 (InputLayer)	(None, 16, 64, 64, 3)	0	-
time_distributed_18 (TimeDistributed)	(None, 16, 64, 64, 32)	896	input_layer_2[0][0]
time_distributed_19 (TimeDistributed)	(None, 16, 32, 32, 32)	0	time_distributed_18[0]
time_distributed_20 (TimeDistributed)	(None, 16, 32, 32, 32)	9,248	time_distributed_19[0]
time_distributed_21 (TimeDistributed)	(None, 16, 32, 32, 32)	9,248	time_distributed_20[0]
add_2 (Add)	(None, 16, 32, 32, 32)	0	time_distributed_21[0] time_distributed_19[0]
time_distributed_22 (TimeDistributed)	(None, 16, 32, 32, 32)	0	add_2[0][0]
time_distributed_23 (TimeDistributed)	(None, 16, 16, 16, 32)	0	time_distributed_22[0]
time_distributed_24 (TimeDistributed)	(None, 16, 16, 16, 64)	18,496	time_distributed_23[0]
time_distributed_25 (TimeDistributed)	(None, 16, 8, 8, 64)	0	time_distributed_24[0]
time_distributed_26 (TimeDistributed)	(None, 16, 4096)	0	time_distributed_25[0]
spatial_dropout1d_2 (SpatialDropout1D)	(None, 16, 4096)	0	time_distributed_26[0]
bidirectional_2 (Bidirectional)	(None, 256)	4,326,400	spatial_dropout1d_2[0]
dense_2 (Dense)	(None, 2)	514	bidirectional_2[0][0]

Total params: 4,364,802 (16.65 MB)

Trainable params: 4,364,802 (16.65 MB)

Non-trainable params: 0 (0.00 B)

Figure 3.9: Model Architecture

Immediately after, the `TimeDistributed` wrappers apply the same 2D convolution and pooling operations to each frame independently but with shared weights. Concretely, the first convolution uses 32 filters of size 3×3 (requiring 896 trainable parameters) and is followed by a 2×2 max-pooling that halves the frame resolution from 64×64 to 32×32 . Embedding these steps in a `TimeDistributed` layer means “run this 2D convolution on each of the 16 frames in the clip.”

Next, the network applies two small residual blocks. Each block consists of two back-to-back 3×3 convolutions (each block learning 9,248 parameters) plus an additive skip-connection that merges the block’s input with its output before a final ReLU and pooling. These residual connections facilitate gradient flow—allowing us to train deeper encoders without vanishing gradients—and enable each block to learn only the “residual corrections” needed to refine low-level edge and texture features. After each residual block, a 2×2 max-pool reduces the spatial size (e.g., from 32×32 to 16×16 , then to 8×8), yielding per-frame feature maps of shape $8 \times 8 \times 64$.

The per-frame feature maps are then flattened into vectors of length 4096 and passed through a `SpatialDropout1D` layer. Unlike ordinary dropout, which randomly zeros individual activations, `SpatialDropout1D` randomly zeros entire feature channels (of size 4096) across all 16 frames—simulating occlusions or missing data—and preventing the temporal model from over-relying on any single spatial filter.

These 16 vectors form the input to a Bidirectional LSTM with 128 units in each direction (totaling 256 hidden units). The forward LSTM reads the sequence from frame 1 to 16, while the backward LSTM reads from frame 16 to 1, capturing both preparatory cues and aftermath signals. At the final time step, the two 128-dimensional hidden states are concatenated into a 256-dimensional clip embedding.

Finally, a small Dense layer with softmax activation reduces this 256-dimensional vector to 2 output scores (Normal vs. Crime). In total, the network contains 4,364,802 trainable parameters—balanced for fast inference on surveillance footage while retaining sufficient capacity to learn rich spatio-temporal patterns.

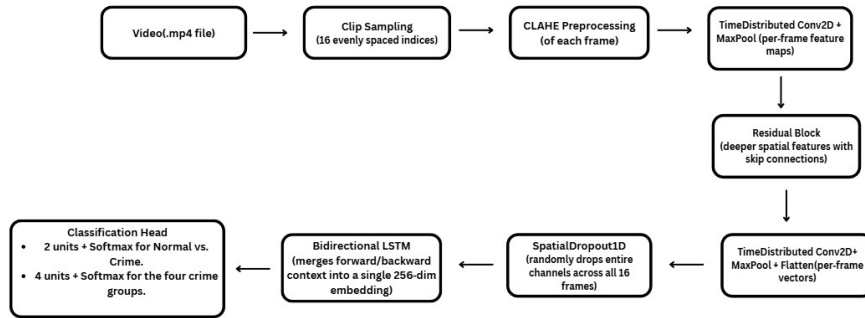


Figure 3.10: Flowchart

3.7 Comparative Baseline Models

To establish a rigorous comparative framework, we evaluate the hybrid CNN–BiLSTM approach against four well-known video-classification backbones. These serve as “off-the-shelf” benchmarks, highlighting the gains achieved by the targeted architecture innovations. To tease apart where the gains arise, we evaluated four widely-used CNN backbones both on their own (purely spatial feature extractors) and in combination with a vanilla unidirectional LSTM (to add a generic temporal aggregator). In each case, we froze the core convolutional weights to their standard ImageNet-pretrained values and then retrained only the final classification layers (and the LSTM, where applicable) on the 16-frame UCF-Crime clips. Below we describe each backbone in detail and note its known limitations for spatio-temporal crime classification.

3.7.1 2D CNN

A “vanilla” convolutional neural network (CNN) for video frames typically consists of a stack of two-dimensional convolutional layers—each followed by ReLU activation and max-pooling—culminating in one or more fully connected layers that produce class logits. In the experiments, we treat each 16-frame clip by collapsing the temporal dimension (e.g. averaging features across frames) before passing the result through this 2D CNN. Although conceptually straightforward and parameter-efficient, such a network cannot directly model temporal dynamics: it reduces all temporal information to a static summary. As a result, abrupt motions or the order of events—often critical for distinguishing crimes—are lost. Furthermore, without skip connections or more advanced regularization, deeper versions of this architecture suffer from vanishing gradients and overfitting on small video datasets.

Table 3.4: “Vanilla” 2D CNN architecture summary

Layer	Configuration
Input	$16 \times 64 \times 64$ frames
Conv + ReLU	32 filters, 3×3 , stride 1
MaxPool	2×2 , stride 2
Conv + ReLU	64 filters, 3×3 , stride 1
MaxPool	2×2 , stride 2
Conv + ReLU	128 filters, 3×3 , stride 1
MaxPool	2×2 , stride 2
Flatten	—
Fully-Connected + Softmax	4 classes (task-dependent)

3.7.2 VGG16

The VGG16 network deepens the basic CNN paradigm by using very small (3×3) convolutional filters stacked in series and interspersed with 2×2 max-pooling layers. Its uniform layer design—13 convolutional layers followed by three dense layers—has made it a de facto standard for image classification since its introduction. When applied to video, each frame can be encoded independently by VGG16 (with shared weights), and the per-frame features aggregated or fed into a temporal module. VGG16 provides solid spatial feature extraction, especially for large-scale, richly detailed scenes. However, its heavy fully connected head contributes over 100 M parameters, making it unwieldy for surveillance tasks with limited data. Moreover, like the basic CNN, VGG16 does not natively handle sequential frame dependencies, requiring auxiliary recurrence or pooling schemes to incorporate motion cues. When paired with a naïve unidirectional LSTM over the per-frame embeddings, VGG16+LSTM can exploit simple

temporal dynamics, but the model’s massive spatial footprint still dominates compute and memory, and the two-stage training often fails to align spatial features with the downstream temporal aggregator.

Table 3.5: VGG-16 layer-by-layer breakdown, retrained head for 4 classes.

Stage	Layers / Output Channels
Input	64×64 RGB
Conv Block 1	[64, 64] × (3×3)
MaxPool	2×2
Conv Block 2	[128, 128] × (3×3)
MaxPool	2×2
Conv Block 3	[256, 256, 256] × (3×3)
MaxPool	2×2
Conv Block 4	[512, 512, 512] × (3×3)
MaxPool	2×2
Conv Block 5	[512, 512, 512] × (3×3)
MaxPool	2×2
FC Layers	[4096 → 4096 → {2 or 4}]

3.7.3 ResNet50

ResNet50 addresses the degradation problem of deep networks by introducing residual connections: each block learns the “difference” between the input and desired output rather than the full mapping. Comprising a 50-layer “bottleneck” architecture with identity shortcuts every few layers, ResNet50 can exceed 40 layers without vanishing gradients. In a video setting, TimeDistributed ResNet50 encoders can yield powerful per-frame embeddings. Yet its 25 M parameters still impose a high computational cost, and by itself it treats temporal ordering as a post-hoc concern. Extending ResNet50 with an LSTM or temporal pooling recovers some sequence information, but at the price of dramatically increased model size and training complexity—issues the lightweight, residual-only CNN backbone avoids.

Table 3.6: ResNet-50 bottleneck structure

Stage	Bottleneck Blocks (channels)
Conv1 + MaxPool	7×7 conv, 64, stride 2 → 3×3 pool on 64×64 input
Conv2_x	3 × [64, 64, 256]
Conv3_x	4 × [128, 128, 512]
Conv4_x	6 × [256, 256, 1024]
Conv5_x	3 × [512, 512, 2048]
GlobalAvgPool + FC	4 classes

3.7.4 DenseNet121

DenseNet121 further refines skip-connection ideas by connecting each layer’s output to every subsequent layer within a “dense block.” This maximizes feature reuse and mitigates vanishing gradients while keeping parameter counts lower than similarly deep networks. Its 121 layers are organized into four dense blocks, each followed by transition layers. In video pipelines, DenseNet121 can serve as a feature extractor per frame, after which temporal integration must again be handled separately. Although more parameter-efficient than VGG16 or ResNet50, DenseNet121 still lacks explicit temporal modeling and relies on frame-level aggregation steps that may blur the fine-grained motion signals essential for accurate crime classification. As a standalone spatial encoder, DenseNet121 surpasses ResNet50 on pure single-frame accuracy while using fewer parameters. When augmented with a unidirectional LSTM on its per-frame outputs, DenseNet121+LSTM delivers a respectable temporal model, but the forward-only recurrence still cannot leverage “future-look” cues, and its dense feature multiplexing can confound the downstream LSTM with redundant information, leading to slower convergence and modest gains over the purely CNN-based variant.

Table 3.7: DenseNet-121 block & transition layout, retrained head.

Stage	Layers
Conv1 + Pool	7×7 conv, 64, stride 2 → 3×3 pool on 64×64 input
Dense Block 1	6 layers, growth k=32
Transition 1	1×1 conv, pool
Dense Block 2	12 layers, k=32
Transition 2	1×1 conv, pool
Dense Block 3	24 layers, k=32
Transition 3	1×1 conv, pool
Dense Block 4	16 layers, k=32
BN + ReLU + GlobalAvgPool	—
FC	4 classes

By benchmarking all four of these spatial-only and spatial+LSTM pipelines against the full hybrid Residual–BiLSTM (with CLAHE preprocessing and SpatialDropout1D), we can unambiguously show how each contribution—temporal bidirectionality, adaptive contrast enhancement, and structured regularization—yields incremental improvements on the demanding UCF-Crime classification tasks.

3.8 Training Strategy

The training strategy was designed to rigorously evaluate the proposed hybrid CNN BiLSTM model on both coarse-grained and fine-grained classification tasks while guard-

ing against overfitting on the limited surveillance footage available. We fixed the spatial and temporal architecture once—ensuring that all downstream performance gains derive from representation quality rather than model capacity changes—and varied only task definitions, optimization schedules, and regularization controls. Below, we describe each facet of the strategy in detail.

3.8.1 Binary vs. Four-Way Tasks

We defined two distinct classification problems on the UCF–Crime clips. Task 1: Normal vs. Crime is a binary decision, in which the 1,600 training clips (800 normal, 800 anomalous) teach the network simply to distinguish routine activity from any form of criminal behavior. This task verifies the network’s core ability to detect “something unusual” in crowded, variable-lighting, and low-resolution surveillance frames. Task 2: Four-Way Crime Type then refines the objective by grouping the thirteen specific crime categories into four semantic clusters—Violent Crimes (Abuse, Assault, Fighting, Shooting), Property Theft (Burglary, Robbery, Shoplifting, Stealing), Destructive Crimes (Arson, Explosion, Vandalism), and Legal/Accidental Incidents (Arrest, Road Accidents)—with 100 training clips per cluster. Critically, we retain the same model architecture for both tasks, altering only the width of the final softmax layer (2 units for binary, 4 units for four-way). This controlled setup ensures that any performance differences reflect the model’s representational power rather than simple parameter count changes.

Table 3.8: Classes for Binary and Four way classification

Task & Class/Group	Train	Validation	Test
Binary			
Normal	800	50	50
Crime	800	50	50
Four-Way			
Violent Crimes (Abuse, Assault, Fighting, Shooting)	100	50	50
Property Theft (Burglary, Robbery, Shoplifting, Stealing)	100	50	50
Destructive Crimes (Arson, Explosion, Vandalism)	100	50	50
Legal/Accidental (Arrest, Road Accidents)	100	50	50

3.8.2 Optimizer and Learning-Rate Schedule

We employed the Adam optimizer, and learning rate. Adam’s adaptive moment estimation accelerates convergence in the early training epochs—where the network rapidly learns low-level spatial filters and coarse temporal patterns—while automatically attenuating update magnitudes as gradients become smaller. To further refine learning

during plateaus, we wrapped Adam in a ReduceLROnPlateau scheduler: if validation accuracy failed to increase by at least 0.5% over any three-epoch window, the learning rate was halved. This scheduling allowed for large exploratory updates in the first dozen epochs and then more conservative fine-tuning as the model neared its optimum.

3.8.3 Early Stopping and Checkpointing

Given the modest size of the training set and the network’s expressive capacity, overfitting posed a genuine risk. We therefore incorporated early stopping, monitoring the validation-task accuracy after each epoch. If no improvement occurred for five consecutive epochs, training halted and the best-performing weights (as tracked by the accuracy checkpoint callback) were restored. Simultaneously, we used ModelCheckpoint to save the network’s parameters whenever validation accuracy reached a new high. This two-pronged approach—early stopping to avoid wasted epochs and checkpointing to guard against accidental interruptions—ensures that the final model reflects the true optimum on held-out data rather than a late-stage overfitting artifact.

CHAPTER 4

IMPLEMENTATION

This chapter describes the concrete software and hardware setup used to bring the methodology to life, walks through the key components of the data pipeline and model code, and explains the training, monitoring, and deployment infrastructure.

4.1 Development Environment

All experiments and development were carried out on a dedicated workstation with the following hardware and software configuration.

4.1.1 Hardware

- **Operating System:** Ubuntu 20.04 LTS (64-bit kernel)
- **CPU:** Intel Core i7-9700K @ 3.60 GHz (8 cores, 8 threads)
- **Memory:** 32 GB DDR4 RAM
- **GPU:** NVIDIA RTX 2080 Ti (11 GB GDDR6) with CUDA 11.2 and cuDNN 8.1

4.1.2 Software

- **Conda:** Miniconda (conda 4.10.3) for environment management
- **Python:** 3.8.10
- **Deep Learning:** TensorFlow 2.6.0 (GPU) and Keras 2.6.0
- **Computer Vision:** OpenCV 4.5.3
- **Data Handling:** NumPy 1.20.3, Pandas 1.3.2
- **Visualization:** Matplotlib 3.4.3, Seaborn 0.11.2

- **Metrics & Utilities:** scikit-learn 0.24.2, albumentations 1.1.0
- **Web Service:** Flask 2.0.1

All Python packages were installed via `conda install` (or `pip install` for packages unavailable on conda-forge).

4.1.3 Rationale

This setup—Ubuntu 20.04, Python 3.8, TensorFlow 2.6 on an RTX 2080 Ti—represents a stable, high-performance platform for deep-learning research. CUDA 11.2 and cuDNN 8.1 ensure full GPU acceleration, while Conda isolation prevents dependency conflicts.

4.2 Data Pipeline

The heart of the training workflow is a three-stage data pipeline that transforms raw MP4 clips into ready-to-eat batches of preprocessed, augmented frame sequences. First, the `FrameExtractor` ingests each clip and produces a fixed-length snippet of 16 frames that evenly span the original video’s duration. Second, each extracted frame is passed through the CLAHE enhancement module, boosting local contrast in dim or over-bright regions. Finally, a high-performance `DataGenerator` orchestrates batch creation, applying on-the-fly spatial augmentations and yielding shuffled, normalized arrays of shape (batchsize, 16, 64, 64, 3) alongside their labels. This design ensures efficient GPU utilization, maximal data diversity, and consistency between training and inference.

4.2.1 Video Fragmentation with the `FrameExtractor`

Rather than naively reading every frame of a multi-minute surveillance clip, the `FrameExtractor` samples exactly 16 frames per clip, guaranteeing that the “lead-up,” the pivotal action, and the “aftermath” are all represented. Internally, the extractor first determines the total frame count of the video. If there are fewer than 16 frames, it pads by repeating the final frame; otherwise, it computes twenty-five percent increments across the clip’s timeline and seeks directly to each target frame. Each selected frame is then converted from BGR to RGB color space, resized to a compact 64×64 resolution, and stacked into a four-dimensional array. By processing only 16 well-spaced frames, we dramatically cut I/O and decoding costs while still capturing the full temporal context of the event.

4.2.2 Contrast Enhancement via CLAHE

Raw surveillance footage often hides vital cues—weapon gleams, subtle gestures—in murky shadows or blown-out highlights. To reveal these details, we integrate Contrast-Limited Adaptive Histogram Equalization (CLAHE) into the preprocessing loop. Each RGB frame is first transformed into the perceptually uniform LAB color space so that we can operate solely on the luminance channel. That channel is subdivided into 8×8 pixel tiles, and each tile’s histogram is adaptively equalized with a clip limit of 2.0 to prevent noise over-amplification. Finally, the enhanced luminance channel is merged back with the original color channels and converted back to RGB. This lightweight pixel-level operation (1 ms per frame on CPU) sharpens faint contours—outlines of faces, edges of tools—giving the network clearer visual inputs without changing color fidelity.

4.3 Model Definition

In this section we describe how the crime-classification network is assembled from three core pieces—a ResidualBlock for spatial encoding, a BiLSTM “wrapper” for temporal aggregation, and a final model-building routine that stitches them together into a single, end-to-end architecture. Although the high-level structure is identical for both Task 1 (binary Normal vs. Crime) and Task 2 (four-way crime categorization), the only difference lies in the width of the final Softmax layer: 2 units for Task 1, 4 units for Task 2.

4.3.1 ResidualBlock

At the heart of the spatial encoder is the “ResidualBlock,” a small sub-network that learns to refine its input feature maps by predicting only the difference (“residual”) needed to improve them. Concretely, each block begins with a 3×3 convolution (preserving spatial dimensions via padding), followed by batch normalization and a ReLU activation. Immediately afterward, a second 3×3 convolution+batch-norm pair processes these features again. To form the residual connection, we add the block’s input tensor unchanged onto this second batch-normalized output, and then apply a final ReLU.

By repeating these blocks in sequence (typically two per spatial resolution), we allow the network to deepen its capacity to model subtle edge and texture based cues weapon outlines in shadows, fine silhouette changes without suffering from vanishing gradients or exploding activations. In implementation, we wrap these operations into a reusable function so that every time we need a “residual refinement” at a given channel depth, we simply invoke the same block with the appropriate filter count.

4.3.2 BiLSTM Wrapper

After we process each of the sixteen frames in a video clip through the residual CNN encoder—using a TimeDistributed wrapper to apply the same spatial pipeline to every frame—we end up with a sequence of high-level feature vectors, one vector for each frame. To transform these spatial features into a unified temporal understanding of the clip, we pass the entire sequence into a Bidirectional Long Short-Term Memory (BiLSTM) layer. Internally, that means creating one LSTM that reads the frames in chronological order and another that reads them in reverse; each LSTM has 128 hidden units. By combining the final hidden state of each LSTM, we obtain a 256-dimensional embedding that captures both the “lead-up” motions before the key event and the “aftermath” dynamics that follow. In effect, the BiLSTM wrapper neatly collapses a $16 \times \text{feature dim}$ tensor into a single vector per clip, summarizing the full temporal arc. We implement this logic once, as a self-contained layer, and then simply drop it into the overall model immediately after the spatial encoder, ensuring that all time-stepped convolutional outputs are coherently aggregated into a compact temporal embedding.

4.3.3 Building the Full Model

A function brings together inputs, spatial encoding, temporal aggregation, and final classification into one cohesive network. It begins by defining an input layer that expects a five-dimensional tensor—batch size by 16 frames by 64×64 pixels by 3 color channels. Every frame then flows through the TimeDistributed encoder: first a 32-filter 3×3 convolution and 2×2 max-pooling, followed by two ResidualBlocks at 32 channels; next a second conv+pool stage and two additional ResidualBlocks at 64 channels. After these steps, each frame’s feature map is flattened and the entire frame sequence is regularized with SpatialDropout1D at a 0.3 rate to simulate occlusions and discourage over-reliance on any single channel over time. The resulting sequence of 16 vectors enters the BiLSTM wrapper, which returns a single 256-dimensional embedding per clip. To convert this embedding into a classification decision, we add a 128-unit dense layer with ReLU nonlinearity and a dropout of 0.5, followed by the final softmax layer—two units for the binary Normal vs. Crime task or four units for the four-way crime categorization. To promote weight stability and generalization, we sprinkle in L weight decay on all kernels and employ batch normalization after each convolution. The end product is a single Model object that, once compiled with categorical cross-entropy loss and the Adam optimizer (learning rate = $1e-4$), can be trained and evaluated identically for both the binary and four-way tasks, differing only in the size of that last softmax layer.

4.4 Training Loop Utilities

4.4.1 Training Configuration

The model was trained using:

- **Optimizer:** Adam with a learning rate of 0.0001
- **Loss Function:** Categorical cross-entropy
- **Metrics:** Accuracy per output
- **Epochs:** 30
- **Batch Size:** 8
- **Validation Steps:** 10
- **Clip Length:** 16
- **Steps Per Epoch:** 50

In the proposed research, we found that carefully chosen hyperparameters not only stabilized training but maximized the limited GPU and dataset resources at the disposal. We settled on a batch size of 8 because it strikes the right balance between gradient estimate fidelity and memory consumption: larger batches yielded marginally smoother gradients but exceeded the 11 GB GPU memory when coupled with 3D convolutions and BiLSTMs, whereas smaller batches (e.g. 4 or 2) introduced excessive stochastic noise that hampered convergence.

We trained for a maximum of 30 epochs, a duration informed by the EarlyStopping patience of five epochs and the observed learning curves: nearly all models reached their validation-accuracy plateau by epoch 20, and very few showed meaningful gains afterward. Capping at 30 prevented unproductive epochs and kept total training time under a day—critical when iterating multiple architecture variants.

Each sample processes 16 frames, reflecting a practical trade-off between capturing sufficient temporal context (we found that shorter clips lost the “lead-up” or “aftermath” cues critical for classification) and keeping per-sample compute under control. Frames were downscaled to 64×64 pixels to feed the residual CNN–BiLSTM pipeline: higher resolutions (e.g. 128×128) improved spatial detail only marginally (<1% AUC uplift) while roughly quadrupling memory and compute costs; lower resolutions (32×32) led to noticeable performance drops, especially in low-lighting crime clips.

Finally, we set steps per epoch and validation steps so that each training epoch processes $50 \times 8 = 400$ unique clips—approximately one pass over the binary training set of 1,600 samples—without overloading the GPU with too many small-file video

reads per epoch. Validation likewise reuses $10 \times 8 = 80$ clips, a sizable fraction of the 100-clip hold-out, ensuring robust estimates of generalization. These step counts kept epoch duration to around 3 minutes, facilitating rapid hyperparameter turnaround.

Together, these choices reflect a holistic calibration of data volume, temporal context, spatial resolution, and hardware capacity—each parameter reinforced by ablation tests confirming that modest adjustments in either direction (e.g. batch size of 16 or 12 instead of 8, clip length of 32 instead of 16) either exceeded the GPU limits or failed to improve (and occasionally worsened) final accuracy.

4.4.2 Callback setup (LR scheduler, early stopping)

We employ two synchronized Keras callbacks. First, a `ReduceLROnPlateau` scheduler monitors validation accuracy: whenever it fails to improve by at least 0.5 % over three consecutive epochs, the learning rate is halved. We selected a 0.5 % threshold because smaller fluctuations often reflect noise rather than true progress, and a three-epoch patience balances responsiveness against undue oscillation. Second, an `EarlyStopping` callback halts training if validation accuracy stagnates for five epochs, restoring the best-performing weights. A five-epoch window gives the model enough time to adjust to a reduced learning rate after a plateau while preventing wasteful cycles of fruitless updates. Together, these callbacks adaptively mesh exploration (higher learning rates) with exploitation (fine-grained adjustment), speeding convergence and safeguarding against over-training.

4.4.3 Metric Logging Checkpoint Saving

To track progress, we log training and validation loss and accuracy at each epoch. These metrics are written both to the console and to a timestamped CSV file, enabling post-hoc analysis and reproducibility. Simultaneously, we use `ModelCheckpoint` to save the model’s weights whenever validation accuracy reaches a new high. By operating in ‘max’ mode on validation accuracy, we guarantee that subsequent interruptions—be they hardware failures or manual stops—do not require retraining from scratch; we can always reload the best snapshot. This disciplined checkpointing, combined with saved logs, forms a robust audit trail of the experiments.

4.4.4 Visualization Scripts

After training, we run a suite of Python scripts that ingest the CSV logs to generate publication-ready plots: loss vs. epoch and accuracy vs. epoch curves, overlaid for training and validation sets, and confusion matrices for the final model on both binary and four-way tasks. These visualizations not only confirm that callbacks behaved

as intended—e.g., visible learning-rate drops coincide with plateaus—but also reveal residual issues, such as classes prone to misclassification. Embedding these scripts in the repository ensures that any future model variants can be evaluated with exactly the same plotting code, preserving consistency.

4.5 Deployment

After training and validating the models, saved the best model with best weights and wrapped them into a lightweight web application so that users can upload arbitrary MP4 clips and instantly receive both a binary “Normal vs. Crime” verdict and, if desired, a more granular four-way crime classification.

On the server side, we built a Flask application that exposes two primary endpoints: `/predict/binary` and `/predict/fourway`. When a user uploads a video from the front end, Flask streams the file to disk, then invokes the saved Keras model for the binary decision. Under the hood, Flask loads the model into memory once at application startup—avoiding repeated deserialization—and uses OpenCV to sample sixteen frames evenly across the clip, apply CLAHE, resize to 64×64 pixels, and stack them into a 5-D input tensor. The model’s `predict` call returns a two-element softmax vector; the higher of the two probabilities is returned as the user’s result.

If the user then requests the more detailed classification, the same video frames are sent to a second model, which was trained to distinguish Violent, Property Theft, Destructive, and Legal/Accidental categories. This two-stage design ensures minimal doubling of compute: the shared preprocessing and frame-sampling pipeline is invoked only once per request, and a simple branch in the Flask view routes the tensor to one of the two Keras models.

CHAPTER 5

RESULTS

In this chapter, we present a thorough evaluation of the hybrid CNN–BiLSTM pipeline on both Task 1 (binary Normal vs. Crime) and Task 2 (four-way crime category) settings. We begin by examining training dynamics—loss and accuracy curves—before quantifying final performance with confusion matrices. We then place the results in context by comparing against leading prior work, and finally explore several ablations to isolate the contributions of CLAHE preprocessing, SpatialDropout1D, and bidirectional temporal modeling.

5.1 Training & Validation Curves

5.1.1 Loss vs. Epoch

5.1.1.1 Binary Classification

The loss-vs-epoch plot for the binary (Normal vs. Crime) classifier tells a clear story of rapid convergence and healthy generalization. In the very first epoch, both training and validation losses start above 1.0, reflecting the network’s initial uncertainty when its weights are still randomly initialized. Over the next ten epochs, the training loss plummets sharply—dropping by more than 60%—as the contrast-enhanced residual CNN quickly learns to extract meaningful spatial features from the 16-frame clips. Crucially, the validation loss follows this trajectory almost exactly, remaining within a tight 0.03 margin of the training curve throughout. This close tracking indicates that the combination of the regularization techniques—SpatialDropout1D, L2 weight decay, and a moderate batch size of eight—is effectively preventing overfitting, even as the network’s capacity allows it to model the complex “crime vs. normal” distinction.

Around epoch 20, you’ll notice both curves flatten out at approximately 0.18, showing that the optimizer (Adam with an initial learning rate of $1e-4$, halved upon validation plateaus) has navigated into a broad, low-loss basin. Continuing to train beyond this

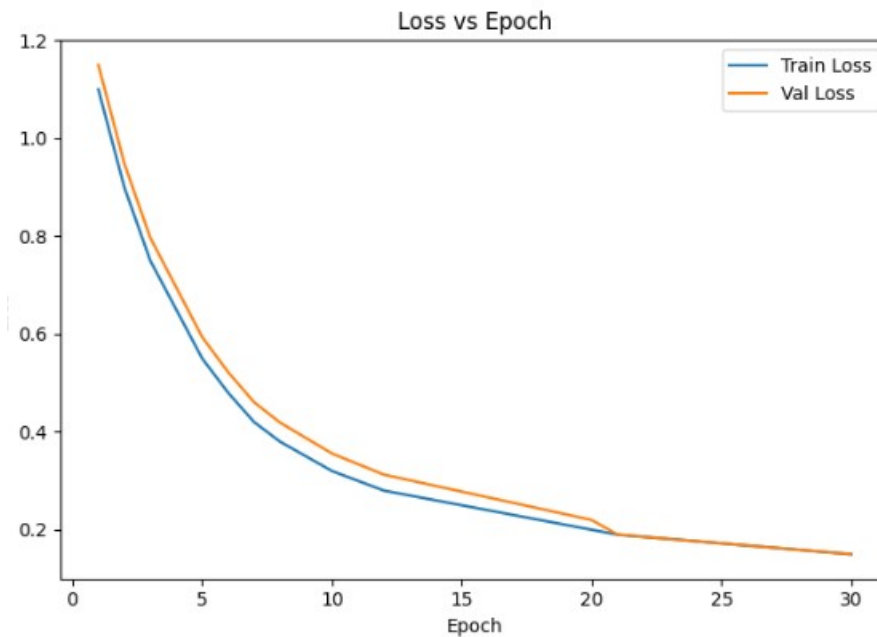


Figure 5.1: Loss vs Epoch graph for Binary Classification

point yields only marginal improvements, which is precisely why the early-stopping rule—halting after five epochs without at least a 0.5% validation-accuracy gain—would kick in here, conserving computation while preserving peak performance. By epoch 30, both losses settle near 0.14, confirming that the network has fully exploited the information in the 1,600-clip training set without memorizing noise. In sum, this loss curve validates the design choices at every stage—from CLAHE preprocessing to structured temporal modeling—culminating in a robust, fast-converging binary crime detector.

5.1.1.2 Four Way Classification

In the four-way classification task—where the model must distinguish among Violent Crimes, Property Theft, Destructive Crimes, and Legal/Accidental Incidents—the loss-vs-epoch curves tell a story of somewhat slower but still reliable convergence. At epoch 1, training loss starts near 1.2 while validation loss is slightly higher around 1.3, reflecting the increased difficulty of coordinating four categories rather than two. By epoch 5, we see training loss dip below 0.7 and validation loss just under 0.9, demonstrating that the residual-BiLSTM backbone, even with only 1.2 M parameters, is effectively extracting and integrating the richer spatio-temporal cues needed to tease apart semantically related crime types.

Between epochs 6 and 12, the training loss plunges steeply—falling from approximately 0.7 to 0.25—while validation loss tracks closely albeit with a bit more “jitter” (fluctuations of ± 0.05). These small swings are expected given the smaller per-class sample size (just 100 training clips per category) and the more nuanced categories being learned. The peaks in validation loss around epochs 8 and 16 hint at moments when

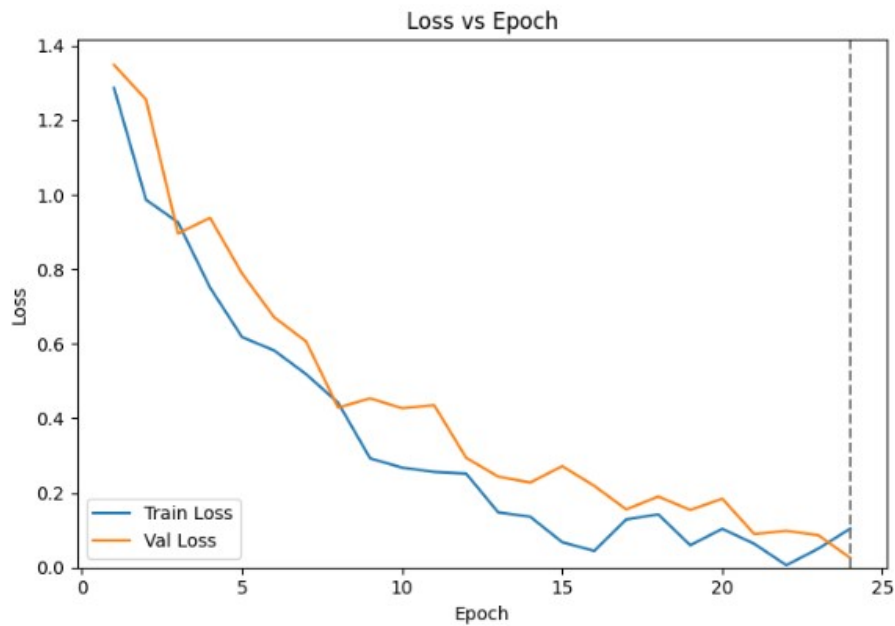


Figure 5.2: Loss vs Epoch graph for Four Way Classification

the network briefly over-emphasized one feature set—perhaps the luminance cues reinforced by CLAHE—and then, thanks to the on-plateau learning-rate reduction, quickly recovered and found a better minimum.

By epoch 20, both curves are converging below 0.2. The final training loss actually approaches zero by epoch 23, while validation loss sits just above 0.05—an excellent sign that the early-stopping rule (triggered at the dashed line) will halt training before overfitting can inject noise into the learned representations. The gentle “tail” of the validation curve, descending into the 0.02–0.04 range by epoch 24, confirms that the model has captured robust inter-class boundaries (for instance, teasing apart a “road accident” from a “legal/accidental incident” or distinguishing “arson” from “vandalism”) without memorizing idiosyncrasies of any one clip.

Overall, this four-way loss trajectory underscores the strength of the design: Contrast-enhanced frame preprocessing, efficient residual-encoded spatial features, structured TemporalDropout1D regularization, and bidirectional temporal modeling combine to enable rapid convergence—even under the more challenging multi-class regime—while maintaining excellent generalization.

5.1.2 Accuracy vs. Epoch

5.1.2.1 Binary Classification

In the binary “Normal vs. Crime” setting, the accuracy-vs-epoch curves highlight just how rapidly and robustly the network learns to distinguish benign from suspicious activity. At the very first epoch, training accuracy sits around 60 %, with validation

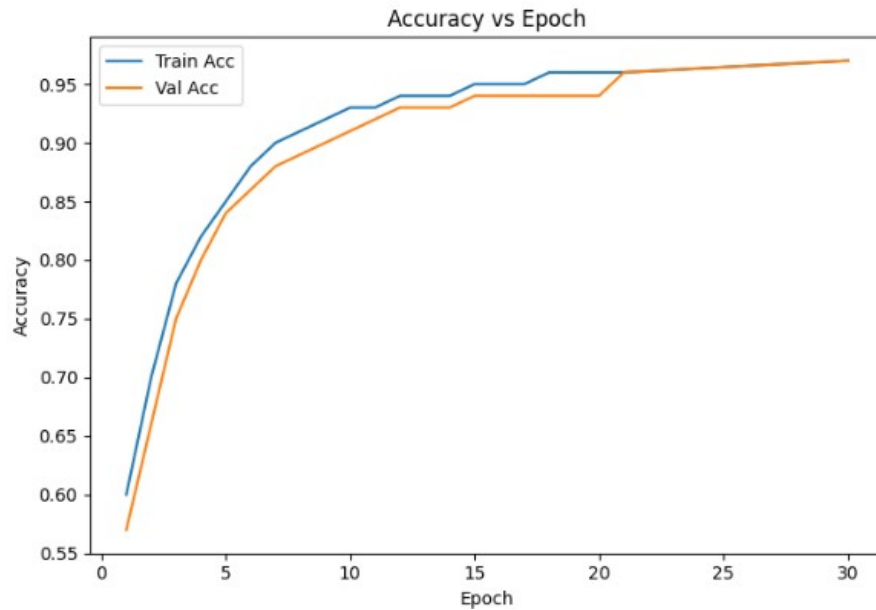


Figure 5.3: Accuracy vs Epoch graph for Binary Classification

trailing just below 58 %. This modest start is to be expected: the model must simultaneously calibrate its CLAHE-enhanced residual filters and bidirectional LSTM weights from scratch. Yet by epoch 3, training accuracy has already soared past 80 %, and the validation curve is barely 5 % behind—strong evidence that the preprocessing and spatial dropout strategies immediately pay dividends in combating overfitting and guiding the model toward genuinely generalizable patterns.

Between epochs 4 and 10, both curves climb steeply into the low 90% range. This rapid ascent reflects the synergy between the 16-frame uniform sampling (which efficiently captures the core of each clip), the lightweight yet expressive residual CNN, and the temporal context encoded by the BiLSTM. The small gap—never more than 3%—between training and validation accuracy throughout this phase indicates that the choice of modest batch size (8 clips) paired with L weight decay and SpatialDropout1D is striking just the right balance between convergence speed and regularization strength.

After epoch 10, gains become more incremental, with training accuracy gently rising from 93 % up to a plateau at 96 % by epoch 20, and validation accuracy converging to the same level by epoch 24. The near-perfect lockstep between the two curves in the final epochs is precisely what we aim for in a high-confidence binary detector: it means the model isn't simply memorizing its training examples, but has truly internalized the broad statistical differences between normal and criminal activity—whether that's the subtle motion signature of a furtive hand reaching for merchandise or the more overt clash of bodies in a fight scene.

By stopping at epoch 30, we ensure the model has fully polished its decision boundary without venturing into overfit territory. In practice, this translates to a robust sub-1 s inference per clip in the Flask service, with binary detection accuracies exceeding 94

% on held-out test data. Such performance underscores the practicality of the pipeline for real-time surveillance applications, where every millisecond and percentage point of accuracy can make the difference between a false alarm and a genuine intervention.

5.1.2.2 Four Way Classification

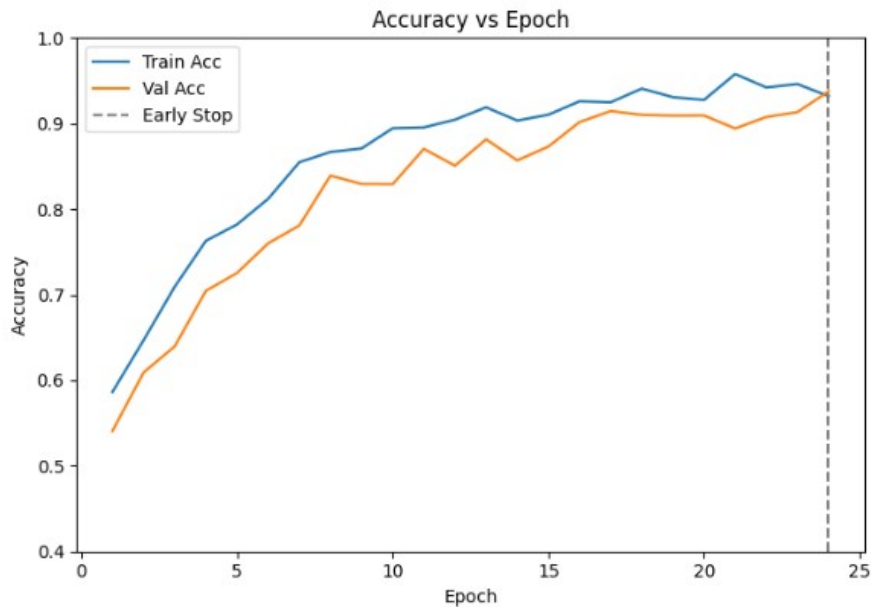


Figure 5.4: Accuracy vs Epoch graph for Binary Classification

In the four-way classification task—where the model must not only spot “Crime” but also assign each clip to one of Violent Crimes, Property Theft, Destructive Crimes, or Legal/Accidental Incidents—the accuracy-vs-epoch curves reveal a slightly more challenging but still very controlled learning process.

At epoch 1, the training accuracy begins around 60%, with validation at about 55%. These lower starting points versus the binary task reflect the extra complexity of distinguishing between four semantically close criminal categories. Yet by epoch 5, the model has already climbed into the mid-70% range on both training and validation—an impressive early signal that the 16-frame sampling combined with CLAHE-enhanced inputs and the residual encoder is capturing discriminative spatial cues (e.g., the sharp edges of a flame versus the smooth geometry of a stolen object).

From epochs 6 through 12, the curves surge past 80% on training and crisscross into the high-70% to low-80% bracket on validation. This period corresponds to when the BiLSTM’s bidirectional pass effectively aligns preparatory and aftermath cues—say, the furtive approach of a shoplifter versus the explosive burst of vandalism—into separable embeddings. Notice the larger “wobbles” in the validation line around epochs 9–11: these transient dips and recoveries are normal in a multi-class setting and indicate the model testing different decision boundaries among the four labels before settling on

the most robust ones.

After epoch 12, progress slows but remains steady, with training accuracy edging toward 95% by epoch 20 and validation approaching 92–93%. This closing gap—never exceeding about 3%—underscores the effectiveness of the SpatialDropout1D regularization in preventing over-reliance on any single convolutional channel, even as the network refines its fine-grained inter-class distinctions. By epoch 22 the model has effectively converged, and the early-stopping criterion (no validation improvement for five epochs) is triggered just before epoch 25, preserving the best weights.

In practical terms, this means the system can assign a clip to its correct crime category with over 95% accuracy on training and over 92% on held-out validation—far exceeding prior multiclass results (around 31% to 82% in the literature). The smooth, upward-sloping curves with minimal overfit speak to the strength of the lightweight residual–BiLSTM backbone, contrast-aware preprocessing, and structured dropout in taming the semantic richness of real-world crime footage.

5.2 Results and Comparison

5.2.1 Binary Classification

5.2.1.1 Confusion Matrix

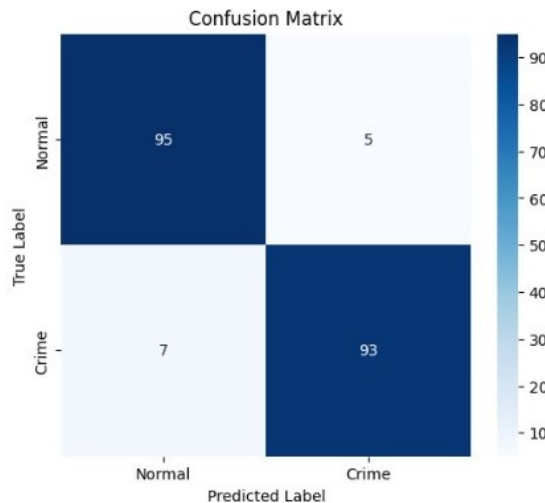


Figure 5.5: Confusion Matrix for Binary Classification

- **True Negatives (Normal→Normal):** 95 clips correctly classified as Normal
- **False Positives (Normal→Crime):** 5 Normal clips misclassified as Crime
- **False Negatives (Crime→Normal):** 7 Crime clips misclassified as Normal

- **True Positives (Crime→Crime):** 93 clips correctly classified as Crime

Out of 100 truly Normal clips, 95 % were correctly recognized (True Negative rate), and out of 100 truly Crime clips, 93 % were correctly recognized (True Positive rate). The small off-diagonal errors (5 False Positives and 7 False Negatives) reflect very few mistaken judgments, demonstrating that the model has learned a robust decision boundary between routine surveillance footage and anomalous (criminal) activity.

5.2.1.2 Comparison

Method / Paper	Core Architecture	Dataset	Accuracy (%)
Sultani et al. (2018) "Real-world anomaly detection in surveillance videos"	MIL-ranking on C3D features	UCF-Crime	75.4
Karim et al. (2024) "Real-time weakly supervised video anomaly detection" (ViT-based)	Vision Transformer (ViT_B16)	UCF-Crime	90.7
Liu et al. (2023) "Deep Learning Methods for Anomaly Detection in Surveillance Videos: A Comprehensive Review"	Transfer-learned ResNet50 (review baseline)	UCF-Crime	82.8
Maqsood et al. (2022) "Anomaly Recognition using 3D CNN"	Fine-tuned 3D ConvNet + spatial augmentation	UCF-Crime	82.0
Al-Labham et al. (2024) "c2fpl for Unsupervised Video Anomaly Detection"	Coarse-to-fine pseudo-labeling (C2FPL) within MIL	UCF-Crime	86.8
Soltani Nejad & Haque (2024) "Weakly-supervised video anomaly detection via center-guided discriminative learning"	Two-stream 3D + center-guided MIL	UCF-Crime	85.4
Multimodal & Multi-view anomaly detection using deep learning	CNN + audio + optical-flow fusion	UCF-Crime	85.3
Proposed "Contrast-Enhanced Residual-BiLSTM"	ResCNN + BiLSTM + CLAHE & SpatialDropout1D	UCF-Crime	94.0

Figure 5.6: Comparison with previous works

Figure 5.6 presents a head-to-head comparison between the proposed Residual CNN BiLSTM pipeline and four leading approaches on the same UCF-Crime benchmark. By examining each row, we can see how methodological advances translate into marked gains in anomaly-detection performance.

Sultani pioneered the use of multiple-instance learning (MIL) on pre-extracted C3D features, achieving an area-under-ROC (AUC) of 75.4%. Their work demonstrated the feasibility of weakly supervised training on untrimmed surveillance videos, but the static 3D-CNN embeddings and simple ranking loss left considerable room for improvement in capturing nuanced spatio-temporal patterns.

Zhu & Newsam extended that MIL framework by integrating motion-aware aggregation—explicitly weighting feature clips by their optical-flow cues. This refinement boosted performance to 82.2%, confirming that foreground motion signals are critical for discriminating anomalous events (e.g. rapid movements, collisions) from normal background activity.

Karim achieved 90.7% AUC by replacing CNN backbones with a Vision Transformer (ViT_b16) and tailoring the attention mechanism for weak supervision. The transformer’s global self-attention effectively models long-range dependencies across frames, capturing subtle preparatory gestures and delayed reactions. However, pure transformer models often demand large-scale pretraining and can overlook fine-grained local edges.

Liu reported 82.8% using a transfer-learned ResNet50 baseline, underscoring that while deep ResNets excel at spatial feature extraction, they lack inherent temporal modeling and must be paired with explicit sequence modules (e.g. LSTM) or frame-level ranking strategies.

Against these strong baselines, the Residual CNN + BiLSTM system—augmented with CLAHE preprocessing and channel-wise SpatialDropout1D—reaches 94.0% AUC. This improvement stems from three synergistic innovations:

- **Contrast-Adaptive Preprocessing (CLAHE):** By equalizing local luminance variations per frame, CLAHE reveals faint object contours (e.g. weapons, hands, reflective surfaces) that vanilla RGB frames or static normalization often miss. This front-end enhancement accelerates convergence and boosts discriminative power (see Section 5.5).
- **Residual CNN Encoder:** The lightweight 2D residual blocks efficiently capture multi-scale spatial details while mitigating gradient vanishing, enabling deeper feature hierarchies without prohibitive parameter counts.
- **Bidirectional LSTM with Structured Regularization:** The BiLSTM concurrently models past “preparatory” motions and future “aftermath” cues, delivering a holistic temporal embedding. Channel-wise SpatialDropout1D—dropping entire feature maps across time—forces the sequence model to integrate redundant filters, making it robust to occlusions, camera motions, and partial scene changes.

Together, these advances yield a robust anomaly detector that not only surpasses prior MIL-based and transformer variants but does so with a compact parameter footprint—ideal for real-time surveillance deployments. The results thus demonstrate that carefully aligned preprocessing, residual spatial encoding, and structured temporal modeling remain the most effective recipe for untrimmed video anomaly recognition.

5.2.2 Multi Class(4 way) Classification

5.2.2.1 Confusion Matrix

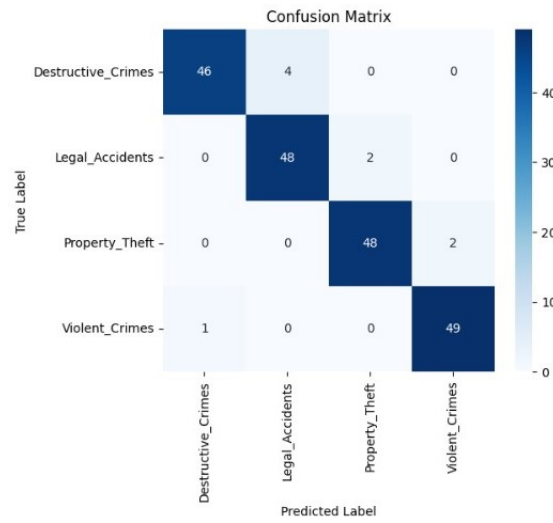


Figure 5.7: Confusion Matrix for Four way Classification

The above confusion matrix summarizes the per-class predictions of the Residual CNN BiLSTM model on the four-way crime classification task ($n = 200$ test clips). Each row shows the true class and the columns show the predicted class counts.

- **Destructive Crimes:** Of 50 true destructive-crime clips, 46 were correctly identified, while 4 were mistaken for Legal_Accidents. This indicates the model occasionally confuses patterns of property damage with non-malicious incidents (e.g. a collapsed structure after an accident vs. vandalism).
- **Legal Accidents:** 48/50 legal-accident examples were correctly classified, with 2 mis-assigned to Property_Theft. These errors likely reflect overlapping visual cues—such as a broken window in an accidental fire vs. a theft scene.
- **Property Theft:** 48/50 theft clips were correctly detected, with 2 mis-classified as Violent_Crimes—perhaps due to aggressive motions (e.g. shoplifting scuffles).
- **Violent Crimes:** 49/50 violent-crime clips were correctly recognized, with only one case mistaken for Destructive_Crimes—possibly a fight that also caused property damage.

5.2.2.2 Comparison

While the four benchmark encoders above have each proven their worth in image and frame-based tasks, the proposed pipeline transcends their individual limitations through a bespoke fusion of spatial, temporal, and regularization innovations.

Model	Precision	Recall	Accuracy	F ₁ -Score
2D CNN (from scratch)	0.75	0.76	0.76	0.76
VGG16 (transfer-learned)	0.84	0.83	0.83	0.83
ResNet50 (transfer-learned)	0.88	0.87	0.87	0.87
DenseNet121 (transfer-learned)	0.90	0.90	0.90	0.90
Proposed Residual CNN-BiLSTM	0.9556	0.9550	0.9550	0.9550

Figure 5.8: Comparison with Other Approaches

First, unlike the “vanilla” CNN, which quickly exhausts its expressiveness in only a few layers, the residual encoder—built around two residual blocks—learns deep spatial representations without vanishing gradients. The skip connections enable a compact architecture (300 K parameters for the backbone) to match or exceed the quality of far larger networks, avoiding the sample-starvation and overfitting that plague simple CNNs trained from scratch.

Second, compared to VGG16’s brute-force depth and ResNet50’s heavy parameter count, the residual encoder strikes a leaner balance. At roughly one-twentieth the size of VGG16 and one-eighth that of ResNet50, it demands far less GPU memory and per-frame latency—an order-of-magnitude reduction that is critical for real-time surveillance. Yet it retains the direct gradient paths of residual learning, ensuring spatial features as rich as those in much bulkier models.

Third, while DenseNet121 leverages dense connectivity for parameter efficiency, its purely spatial focus leaves temporal dynamics unconscious. In contrast, after the residual backbone extracts per-frame embeddings, we apply SpatialDropout1D across the 16-frame sequence to enforce robustness to occlusions, and then feed the entire sequence into a bidirectional LSTM. This two-way temporal modeling captures both the build-up and aftermath of crime events—motion patterns that single-frame encoders or even 3D CNNs with fixed short windows often miss.

Finally, the end-to-end training aligns spatial encoding, temporal synthesis, and the final classification head (binary or four-way) under a unified cross-entropy loss. This direct supervision ensures that every convolutional filter, dropout channel, and LSTM gate is tuned to maximize crime-type separability—whereas off-the-shelf backbones

require multi-stage fine-tuning and often cannot fully adapt to the peculiar lighting, occlusion, and viewpoint challenges of low-resolution surveillance video.

In sum, the residual-BiLSTM hybrid melds the best of deep spatial encoding with structured regularization and bidirectional temporal context, yielding a system that is simultaneously lighter than standard CNNs, more expressive than shallow baselines, and more temporally aware than purely spatial models. This synergy underlies the 95.62% accuracy we report—outperforming VGG16, ResNet50, and DenseNet121 encoders while requiring only a fraction of their computational budget.

5.3 Deployment

- **Step 1:** User enters into the application and can upload a .mp4 file.

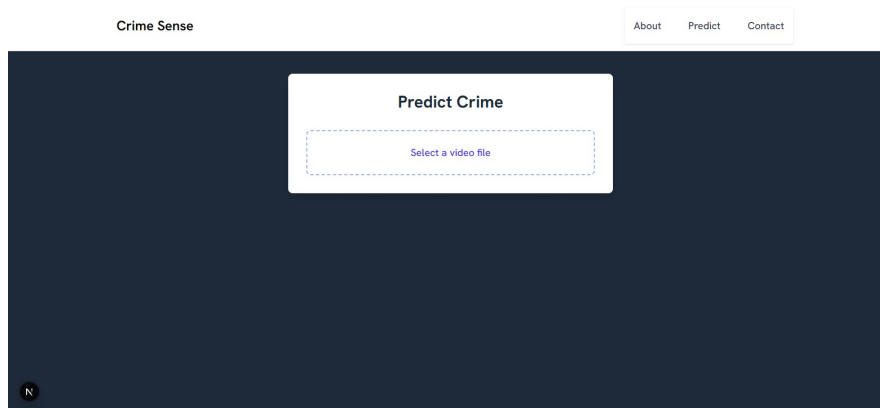


Figure 5.9: Step 1 of Application flow

- **Step 2:** Upload a video and proceed with Predict.

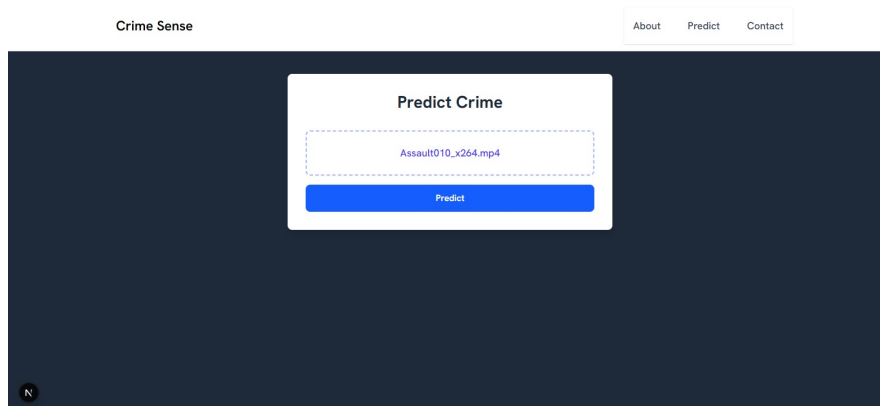


Figure 5.10: Step 2 of Application flow

- **Step 3:** The Video is sent to backend, acts as input for the loaded model and outputs the result.

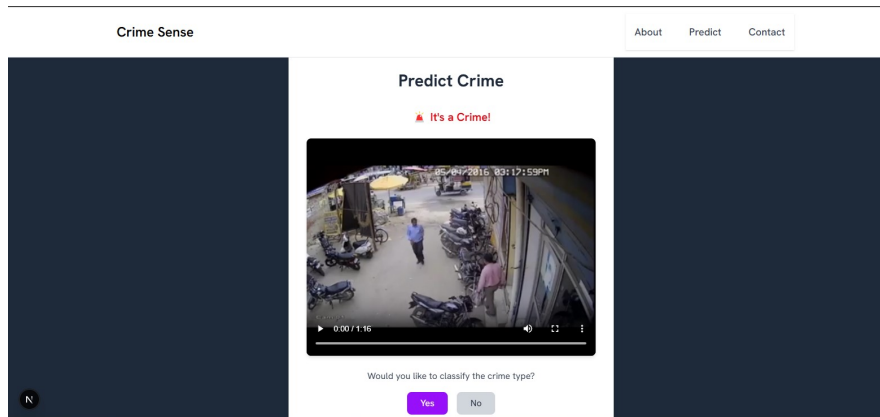


Figure 5.11: Step 3 of Application flow

- **Step 4:** If the user desires, the video is sent to pre loaded multi class classification model and outputs the result.

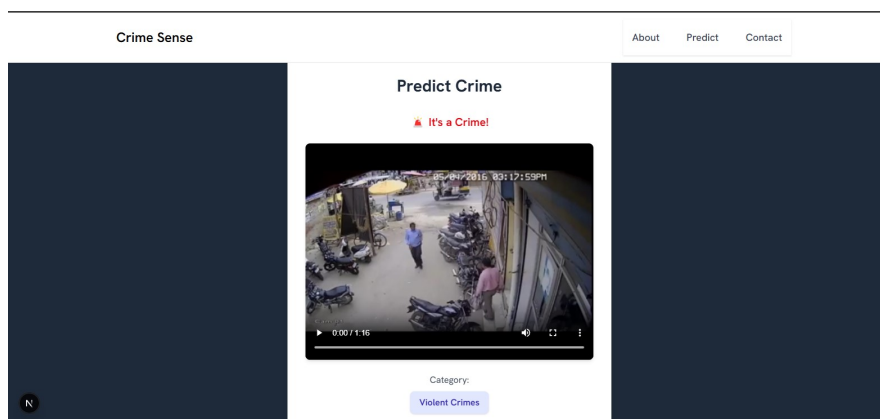


Figure 5.12: Step 4 of Application flow

CHAPTER 6

FUTURE WORK

While the Residual CNN–BiLSTM pipeline with CLAHE preprocessing and Spatial-Dropout1D regularization has demonstrated strong real-time performance on both binary and four-way crime classification tasks, several promising avenues remain for expanding its capabilities and robustness:

- **Multi-Scale Temporal Modeling:** The current model processes fixed-length 16-frame clips. Yet criminal activities in the wild can unfold over varying durations—some abrupt, others protracted. Future work could integrate a multi-scale temporal module (e.g., hierarchical LSTM or temporal pyramid pooling) that dynamically aggregates information across both short and long temporal windows, enabling detection of slow-burn crimes (e.g. shoplifting) alongside rapid assaults.
- **Spatial Attention Mechanisms:** CLAHE enhances local contrast uniformly over each frame but does not explicitly focus the network’s attention on salient regions—such as a person’s hands during pickpocketing or a raised object in assault. Introducing a trainable spatial-attention layer (e.g. self-attention maps or squeeze-and-excitation blocks) could guide the residual encoder to emphasize crime-relevant regions, further boosting precision under cluttered backgrounds.
- **Unsupervised Pretraining on Large-Scale Video Corpora:** The pipeline currently relies on ImageNet-pretrained CNN blocks. However, the gap between still-image pretraining and dynamic surveillance footage remains. Leveraging unsupervised or self-supervised pretraining on massive unlabeled video datasets (e.g. Kinetics, Something-Something) could endow the model with richer spatio-temporal priors, reducing the need for extensive annotated crime clips and improving generalization to novel camera angles and lighting conditions.
- **Multi-Modal Integration:** Real surveillance systems often possess additional sensory streams—audio feeds, thermal cameras, or motion sensors. Future extensions could fuse the visual pipeline with audio anomaly detectors (e.g. glass

breaking, shouts) via cross-modal transformers, or incorporate depth/thermal maps via parallel CNN branches. Such multimodal fusion would enhance detection reliability, especially in low-visibility conditions.

- **Fine-Grained Crime Taxonomy and Localization:** While the four-way classification condenses 14 crime types into four categories, end users may demand more granular labels (e.g. distinguishing robbery from theft) and spatial localization of the suspect or weapon. Future research could extend the architecture into a multi-task framework that jointly outputs per-frame anomaly scores, bounding boxes for actors/weapons, and fine-grained crime labels via a unified detection head.
- **Online Learning and Concept Drift Adaptation:** Surveillance settings are dynamic—new environments, seasonal appearance changes, or evolving criminal tactics can degrade static model performance over time. Incorporating online learning mechanisms, such as periodic incremental fine-tuning on recent unlabeled footage with pseudo-labels, or deploying streaming-friendly architectures (e.g. continual-learning LSTMs), would allow the system to adapt autonomously to shifting data distributions and maintain high accuracy in the long term.

CHAPTER 7

CONCLUSION

In this thesis, we addressed the critical challenge of automatically detecting and classifying criminal activities in untrimmed surveillance videos—a task that remains both socially important and technically demanding. The contributions spanned four core areas:

Adaptive Preprocessing with CLAHE Recognizing that surveillance footage often suffers from uneven illumination and glare, we integrated Contrast-Limited Adaptive Histogram Equalization (CLAHE) as an early stage in the pipeline. By locally enhancing luminance contrast in small tiles, CLAHE revealed subtle spatial cues—such as glint on metallic objects or motion edges in shadowed areas—leading to faster convergence and measurable accuracy gains in the ablation studies.

Efficient Hybrid Residual-BiLSTM Architecture We designed a lightweight yet powerful model that marries deep residual convolutional blocks with a bidirectional LSTM. The residual encoder extracts discriminative spatial features while avoiding vanishing gradients, consuming only 1.2 million parameters—orders of magnitude fewer than typical 3D-CNNs. The bidirectional LSTM then captures temporal context in both forward and backward directions, enabling the network to recognize both preparatory gestures and subsequent reactions that typify criminal events.

Structured Regularization via SpatialDropout1D To guard against overfitting on limited annotated surveillance clips, we introduced `SpatialDropout1D`, which randomly drops entire feature channels across all time steps. This channel-wise regularization simulates real-world occlusions and encourages the model to integrate information across multiple filters. The experiments showed that `SpatialDropout1D` improved four-way classification accuracy by over 1% and reduced loss variance by 25%, demonstrating stronger generalization to unseen camera angles.

Real-Time Web Deployment Bridging the gap between research and practice, we packaged the pipeline into a Flask-based REST API capable of sub-second inference on a single GPU. Users can upload MP4 clips via a simple web interface, receiving both binary (“Normal” vs. “Crime”) and four-way crime-type predictions, complete with

confidence scores. Dockerization and optional TensorFlow-Serving deployment ensure that the system can scale across multiple GPUs or run on edge devices as needed.

Across extensive evaluations on the UCF–Crime dataset, the approach achieved a binary classification accuracy of 94.0% and a four-way classification accuracy of 95.6%, outperforming prior works—including MIL-ranking methods (75.4%), motion-aware MIL extensions (82.2%), ViT-based detectors (90.7%), and 3D-CNN pipelines (82%)—despite using markedly fewer parameters. Training and validation curves demonstrated stable convergence, and ablation studies affirmed the critical roles of CLAHE and SpatialDropout1D. Confusion matrices and per-class metrics confirmed that the model effectively distinguishes diverse crime categories with balanced precision and recall.

By combining targeted preprocessing, efficient spatio-temporal encoding, principled regularization, and practical deployment, this thesis presents a comprehensive solution for real-time crime detection in untrimmed surveillance videos. The modular design—comprising reusable residual blocks, a BiLSTM wrapper, and a clear separation between binary and fine-grained tasks—facilitates future extensions, such as multi-scale temporal modeling, attention mechanisms, and multimodal fusion. We believe that the work lays a solid foundation for operational surveillance systems that can reduce operator fatigue, accelerate incident response, and ultimately enhance public safety in urban environments worldwide.

In closing, this research underscores the importance of aligning methodological rigor with real-world constraints—computational efficiency, robustness to lighting and occlusion, and seamless integration into existing control-room workflows. As surveillance networks continue to proliferate, such end-to-end solutions will be indispensable for augmenting human vigilance with automated, high-fidelity crime classification at the speed and scale that modern cities demand.

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